RV College of Engineering

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Intelligent Clinical Document Processing and SNOMED Mapping System

MAJOR PROJECT REPORT
(AI481P)

Submitted by,

Aryan Sinha	**1RV22AI009**
Kushagra Aatre	**1RV22AI025**
Nischitha P	**1RV22AI031**
Shreya M	**1RV22AI054**

Under the guidance of

Dr. Somesh Nandi
Associate Professor
Dept. of AIML
RV College of Engineering

In partial fulfillment for the award of degree of

Bachelor of Engineering
in
Artificial Intelligence and Machine Learning

Department of AIML
2025-26

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Go, change the world

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(*Autonomous institution affiliated to VTU, Belagavi*)

Department of Artificial Intelligence and Machine Learning

CERTIFICATE

Certified that the major project (AI481P) work titled ***Intelligent Clinical Document Processing and SNOMED Mapping System*** is carried out by **Aryan Sinha** (**1RV22AI009**), **Kushagra Aatre** (**1RV22AI025**), **Nischitha P** (**1RV22AI031**) and **Shreya M** (**1RV22AI054**) who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in **Artificial Intelligence and Machine Learning** of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project report deposited in the departmental library. The major project report has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said degree.

Guide	Head of the Department	Principal
Dr. Somesh Nandi	Dr. Shantha Rangaswamy	Dr. K. N. Subramanya

External Viva

Name of Examiners	Signature with Date
1.	
2.	

DECLARATION

We, **Aryan Sinha**, **Kushagra Aatre**, **Nischitha P** and **Shreya M** students of eighth semester B.E., Department of Artificial Intelligence and Machine Learning, RV College of Engineering, Bengaluru, hereby declare that the major project titled '**Intelligent Clinical Document Processing and SNOMED Mapping System**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in **Artificial Intelligence and Machine Learning** during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and we will be one of the authors of the same.

Place: Bengaluru

Date:

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1.	Aryan Sinha(1RV22AI009)	
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ABSTRACT

The healthcare sector generates vast quantities of unstructured clinical data through documents such as referral letters and discharge summaries. Manual processing of these records is time-consuming and error-prone, creating significant bottlenecks in clinical workflows. This project addressed the challenge of automated medical document understanding and Systematized Nomenclature of Medicine — Clinical Terms (SNOMED CT) coding within a template-free intelligent extraction framework.

The domain of clinical Natural Language Processing (NLP) has expanded rapidly with the adoption of Electronic Health Records (EHR) and the growing volume of clinical documents requiring structured coding. However, many existing systems rely on rigid, template-dependent approaches that fail to adapt to diverse document layouts commonly encountered in real-world National Health Service (NHS) environments. This limitation often results in incomplete or inaccurate code mappings.

The proposed system, developed as an Intelligent Clinical Document Processing System (ICDPS), aimed to: (i) create an intelligent ingestion module for PDF and image-based clinical documents; (ii) develop a non-template-based extraction engine for diverse layouts; (iii) automate clinical summarization and categorization into predefined medical fields; and (iv) normalize free-text clinical concepts to SNOMED CT codes.

The methodology followed an iterative Software Development Life Cycle (SDLC) consisting of requirements analysis, design, implementation, testing, and deployment. OCR pipelines handled image-based inputs, while direct extraction processed PDFs. Transformer-based NLP models were employed for Named Entity Recognition (NER) and classification of entities such as Diagnosis, Medication, Procedure, and Allergy. A confidence-ranked SNOMED CT suggestion interface assisted users in selecting suitable codes.

The system achieved clinical entity extraction accuracy exceeding 88% and SNOMED CT mapping precision of 85% on the test dataset, satisfying project requirements. The resulting system demonstrated potential to reduce administrative burden, accelerate structured record generation, and improve the quality of medical coding in clinical environments.

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CHAPTER 1

INTRODUCTION

Healthcare organizations generate large volumes of clinical documents every day, including referral letters, discharge summaries, laboratory reports, outpatient records, and clinical notes. These documents contain important information required for patient care, clinical decision-making, and administrative processes. However, much of this information is stored as unstructured text, making it difficult to search, analyse, and integrate into digital healthcare systems.

The increasing adoption of Electronic Health Records (EHRs) has improved the availability of clinical data, but it has also highlighted the challenges associated with processing large collections of heterogeneous documents. Many healthcare workflows still rely on manual review and coding of clinical information, which can be time-consuming and prone to inconsistencies. As the volume of digital health records continues to grow, there is a need for automated methods that can extract clinically relevant information accurately and efficiently.

Recent advances in Natural Language Processing (NLP) and transformer-based language models have significantly improved the ability of computers to understand medical text. At the same time, improvements in Optical Character Recognition (OCR) technologies have enabled the extraction of text from scanned documents and image-based reports. These developments make it possible to design systems that can process clinical documents from different sources and convert them into structured representations suitable for downstream healthcare applications.

To address this problem, this project develops an Intelligent Clinical Document Processing System (ICDPS) capable of processing both PDF and image-based clinical documents. The system combines OCR, clinical Named Entity Recognition (NER), document understanding, and SNOMED CT terminology mapping within a unified processing pipeline. The objective is to automatically extract clinically meaningful information from unstructured documents and represent it in a structured form that can support healthcare information management and clinical coding workflows.

This chapter introduces the problem domain, discusses the motivation and objectives of the work, reviews relevant literature, identifies existing research gaps, and presents the overall methodology adopted in the development of the proposed system.

1.1 Introduction

Processing clinical documents remains a challenging task in modern healthcare environments. Hospitals and healthcare institutions routinely generate referral letters, discharge summaries, diagnostic reports, prescriptions, and other forms of clinical documentation. Although these documents contain valuable patient information, much of the content is recorded as free-text narratives, making automated analysis difficult.

Traditionally, the extraction and coding of clinical information have relied on manual effort from healthcare professionals and clinical coders. While manual review allows human interpretation of complex medical information, it is often time-intensive and can lead to variability in coding outcomes. As healthcare organizations continue to digitize records and expand EHR adoption, the volume of clinical text requiring review has increased considerably.

Advances in artificial intelligence and NLP have created new opportunities for automating parts of this workflow. Transformer-based models such as BioBERT and ClinicalBERT have demonstrated strong performance on clinical information extraction tasks, enabling more accurate identification of diseases, medications, procedures, and other medical concepts from unstructured text.

In addition to textual complexity, healthcare organizations frequently manage documents in multiple formats. Some documents contain selectable digital text, while others are available only as scanned images or PDF files. As a result, effective clinical document processing requires both reliable text extraction and robust language understanding capabilities.

The work presented in this project focuses on developing a template-independent framework for clinical document processing and SNOMED CT coding. The proposed system combines OCR-based text extraction, BioBERT-based entity recognition, clinical information categorization, and semantic SNOMED CT mapping. By integrating these components into a single pipeline, the system aims to transform unstructured clinical narratives into structured information that can support healthcare documentation and coding activities.

1.1.1 Terminology and Definitions

The development of the Intelligent Clinical Document Processing System (ICDPS) involves several concepts and technical terms from healthcare informatics, Natural Language Processing (NLP), Artificial Intelligence (AI), and clinical coding systems. Understanding these terms is essential for interpreting the design and implementation aspects of the proposed system. The important terminology used throughout this report is presented below.

- **Clinical Document:** A healthcare-related document containing patient information, medical history, diagnoses, prescriptions, laboratory reports, referrals, discharge summaries, and clinical observations.
- **Natural Language Processing (NLP):** A branch of Artificial Intelligence concerned with enabling computers to understand, process, analyze, and generate human language.
- **Optical Character Recognition (OCR):** A technology used for converting text embedded within scanned documents or images into machine-readable textual information.
- **Named Entity Recognition (NER):** A Natural Language Processing task used to identify and classify specific entities from text into predefined categories such as diseases, medications, procedures, symptoms, and patient details.
- **SNOMED CT:** Systematized Nomenclature of Medicine – Clinical Terms, a comprehensive clinical terminology system used to represent medical concepts in a standardized manner.
- **Electronic Health Record (EHR):** A digital version of a patient's medical history that includes clinical data collected and maintained across healthcare systems.
- **Transformer Model:** A deep learning architecture that uses attention mechanisms to capture contextual relationships within sequential data and is widely used in modern NLP systems.
- **Intelligent Clinical Document Processing System (ICDPS):** The proposed framework designed to automate extraction of clinical information from heterogeneous healthcare documents and map extracted information to standardized SNOMED CT codes.

1.2 Overview of the Project Topic

Clinical documents contain a large amount of information that is essential for patient care, including diagnoses, medications, laboratory results, procedures, and clinical observations. Although this information is routinely recorded within healthcare systems, much of it remains embedded in unstructured text documents. Extracting clinically meaningful information from these documents is a challenging task that requires techniques from healthcare informatics, natural language processing, and machine learning.

The development of modern NLP models has made it possible to automatically identify medical concepts, classify clinical information, and support standardized clinical coding. These capabilities have increased interest in intelligent clinical document processing systems that can reduce manual effort while improving the accessibility and usability of healthcare data. The following subsections discuss the broader context of the problem, its societal relevance, and the technical challenges that motivated this project.

1.2.1 Global Scenario

Healthcare organizations across the world are increasingly relying on digital health records and electronic documentation systems. As a result, the amount of clinical text generated during routine healthcare activities has grown substantially. Referral letters, discharge summaries, outpatient reports, laboratory reports, and clinical notes are now routinely stored in digital form, creating large repositories of healthcare information.

This growth has encouraged significant research activity in clinical natural language processing. Early systems primarily relied on handcrafted rules and predefined templates to identify medical concepts. While such approaches performed adequately in controlled environments, their performance often decreased when applied to documents with different formats and writing styles. More recently, transformer-based language models have demonstrated improved performance on clinical information extraction tasks by learning contextual relationships directly from text.

Research datasets and benchmark initiatives such as i2b2 have played an important role in evaluating these approaches and encouraging the development of more robust extraction systems. Models including BioBERT [1] and ClinicalBERT [2] have demonstrated strong performance on clinical named entity recognition and related biomedical NLP tasks, making them attractive candidates for healthcare document processing applications.

At the same time, healthcare standards organizations continue to promote the adoption of structured clinical terminologies and interoperable health records. This has increased the importance of systems capable of converting unstructured clinical narratives into standardized representations that can be shared and reused across healthcare environments.

1.2.2 Societal Relevance

The ability to process clinical documents automatically has practical benefits for patients, healthcare professionals, and healthcare organizations.

For patients, accurate extraction and coding of clinical information can improve the consistency of medical records and reduce the likelihood of important information being overlooked during treatment. Structured clinical data also supports continuity of care when information must be exchanged between different healthcare providers.

For healthcare professionals, document review and clinical coding can require significant time and effort, particularly when dealing with large volumes of records. Automating routine extraction tasks can reduce administrative workload and allow clinicians and coders to focus on activities that require expert judgement.

Structured and standardized clinical information is also valuable at an organizational level. Healthcare institutions use coded clinical data for reporting, resource planning, quality monitoring, research activities, and interoperability with external healthcare systems. Terminology standards such as SNOMED CT provide a common representation of medical concepts, enabling consistent interpretation of clinical information across different systems and organizations.

1.2.3 Problem Area

Although substantial progress has been made in clinical NLP, several challenges remain when applying these techniques to real-world clinical documents.

A major challenge is the diversity of document formats encountered in healthcare environments. Clinical information may appear in referral letters, discharge summaries, scanned reports, laboratory documents, or handwritten forms that have been digitized. Systems designed around fixed templates or predefined document structures often struggle to generalize across these variations.

Another challenge arises from the complexity of clinical terminology. Medical concepts are frequently expressed using abbreviations, synonyms, incomplete phrases, and context-dependent language. Correctly identifying the intended meaning of a clinical entity and mapping it to an appropriate SNOMED CT concept requires more than simple keyword matching.

In addition, many existing solutions focus primarily on digitally generated text and provide limited support for image-based documents. In practice, scanned referral letters and legacy clinical records remain common in many healthcare settings. These documents require reliable OCR processing before downstream NLP techniques can be applied.

The combination of document variability, terminology complexity, and the need for standardized coding creates a challenging problem that cannot be addressed effectively through traditional rule-based approaches alone. These limitations motivated the development of the proposed Intelligent Clinical Document Processing System (ICDPS), which integrates OCR, clinical entity extraction, and SNOMED CT mapping within a unified processing framework.

1.3 Specific Details of the Project Topic

The Intelligent Clinical Document Processing System (ICDPS) was developed as a modular pipeline to process clinical documents from ingestion to standardized clinical coding. The overall architecture consists of four major stages: document ingestion, clinical information extraction, summarization and categorization, and SNOMED CT mapping.

The document ingestion stage is responsible for handling different document formats commonly encountered in healthcare environments. Documents containing selectable text are processed directly using the PyMuPDF library, while scanned reports and image-based documents are processed through an OCR pipeline based on Tesseract 5.0. Both approaches generate a normalized text representation that is forwarded to the downstream NLP components.

Clinical information extraction forms the central component of the system. A fine-tuned BioBERT model is used to identify clinically relevant entities from extracted text. The model was trained using the curated dataset developed for this project and supplemented with NHS-style annotations to improve performance on representative clinical documents. Extracted entities are classified into categories such as Diagnosis, Medication, Dosage, Procedure, Anatomy, Allergy, Lab Result, and Temporal Expression.

Following entity extraction, a categorization module assigns identified entities to role-specific action groups to support clinical workflows. For example, medication-related information may be directed towards pharmacist review, while diagnostic findings can be highlighted for clinician attention.

The summarization component generates concise structured summaries by selecting clinically important sentences based on entity density and relevance. The resulting output provides a simplified overview of the document while preserving key clinical information.

To support standardized terminology mapping, the system incorporates a SNOMED CT retrieval and ranking module. Candidate concepts are initially retrieved through semantic similarity search using a FastText-based embedding index of SNOMED CT descriptions. Retrieved concepts are then reranked using contextual information from the source document to generate confidence-ranked coding suggestions.

1.4 Purpose and Motivation

The increasing volume of clinical documentation presents a significant challenge for healthcare organizations that depend on accurate extraction and coding of patient information. Manual review of referral letters, discharge summaries, and clinical reports can be time-consuming, particularly when large numbers of documents must be processed within limited timeframes.

This project was undertaken to investigate whether recent advances in clinical NLP could be applied to automate parts of this workflow. The goal was not to replace clinical judgement, but to provide structured information and coding suggestions that could reduce repetitive manual effort and support faster review of clinical documents.

Another motivation was the need for more consistent mapping of extracted clinical information to standardized medical terminologies. Since multiple SNOMED CT concepts may be relevant for a given clinical phrase, selecting appropriate codes often requires additional effort from healthcare professionals. Providing confidence-ranked suggestions can assist users in identifying suitable concepts more efficiently.

The availability of open-source biomedical language models, publicly accessible research datasets, and established clinical NLP benchmarks also made it possible to explore advanced document processing techniques within an academic project setting. These resources provided a practical foundation for developing and evaluating an end-to-end clinical document processing pipeline.

1.5 Problem Statement

Healthcare organizations generate large amounts of clinical information in the form of referral letters, discharge summaries, laboratory reports, and other narrative documents. Much of this information remains embedded within unstructured text, making automated extraction and standardization difficult.

Existing solutions often depend on predefined templates, document-specific rules, or manually engineered patterns. Such approaches may perform adequately for known document formats but frequently struggle when applied to documents originating from different healthcare organizations or clinical specialties. In addition, many systems provide limited support for scanned and image-based documents, which remain common in healthcare environments.

The problem addressed in this project is the development of a template-independent system capable of processing heterogeneous clinical documents in PDF and image formats, extracting clinically relevant entities, generating structured summaries, and producing confidence-ranked SNOMED CT coding suggestions with minimal manual intervention.

1.6 Objectives

The objectives of the project are:

1.7 Objectives

The objectives of the project are:

1. To design and implement a document ingestion pipeline capable of processing both digital PDF documents and image-based clinical records using direct text extraction and OCR techniques.

2. To develop a BioBERT-based clinical entity extraction module capable of identifying key medical entities across multiple clinical categories.

3. To generate structured summaries of clinical documents and organize extracted information into role-specific review categories.

4. To develop a SNOMED CT mapping module that retrieves and ranks candidate concepts using semantic similarity techniques and contextual relevance scoring.

5. To achieve accurate terminology mapping by providing confidence-ranked SNOMED CT code suggestions for extracted clinical entities.

6. To evaluate the complete end-to-end system using representative clinical documents and standard NLP evaluation metrics.

1.8 Scope and Relevance

This project focuses on the design, implementation, and evaluation of an Intelligent Clinical Document Processing System (ICDPS) for extracting structured information from English-language clinical documents. The system combines document processing, clinical natural language processing, and terminology mapping techniques to transform unstructured clinical text into structured outputs that can be used for downstream healthcare applications.

The implementation supports multiple document formats, including digitally generated PDF files, scanned documents, and image-based records such as JPEG, PNG, and TIFF files. Text is extracted using either direct PDF parsing or OCR, depending on the document type. The extracted content is then processed through a clinical NLP pipeline that performs entity extraction, information categorization, summarization, and SNOMED CT mapping. The final output is generated in a structured machine-readable format, allowing extracted information and coding suggestions to be reviewed by end users before acceptance.

The work is limited to clinical document processing and terminology mapping. The system does not perform medical diagnosis, treatment recommendation, or clinical decision-making. Integration with live Electronic Health Record systems, real-time patient monitoring platforms, and large-scale production healthcare infrastructure is outside the scope of the current implementation. Support for non-English clinical documents and complex handwritten records is also not included in the present version of the system.

The project was developed and evaluated as a prototype in a controlled environment. The current implementation operates using batch document processing, while large-scale deployment, real-time APIs, and enterprise healthcare integration remain potential areas for future development.

The system assumes that input documents are reasonably readable and that scanned records contain sufficient image quality for OCR processing. Performance may be affected by poor scan quality, unusual document layouts, OCR errors, and computational requirements associated with transformer-based NLP models.

The relevance of this work lies in its ability to combine OCR, clinical entity extraction, semantic retrieval, and SNOMED CT mapping within a single processing pipeline. Such systems can help reduce manual review effort, improve the consistency of clinical coding workflows, and increase the accessibility of information contained within unstructured healthcare documents. The techniques explored in this project are also relevant to ongoing research in clinical NLP, healthcare informatics, and intelligent document understanding.

1.9 Literature Review

A review of existing literature was carried out to identify current approaches used in clinical document processing, biomedical natural language processing, information extraction, and clinical terminology mapping. Examining previous research helped establish the technical foundations of the proposed system and provided insight into the strengths and limitations of existing methods.

The following sections summarize relevant studies, compare their methodologies and findings, and highlight the research gaps that motivated the development of the proposed ICDPS framework.

1.9.1 Literature Survey

Table 1.1 summarizes representative research contributions relevant to this work. The reviewed studies cover a range of topics including biomedical language models, clinical named entity recognition, OCR-based document processing, ontology mapping, semantic retrieval, relation extraction, de-identification, explainable AI, and healthcare information systems.

Each study was examined with respect to its methodology, key contributions, advantages, limitations, and relevance to the objectives of this project. The review helped identify techniques suitable for clinical entity extraction, document understanding, and SNOMED CT mapping, while also revealing limitations in existing systems related to document variability, terminology standardization, and support for heterogeneous clinical records.

The observations obtained from the literature survey guided the selection of models, system components, and architectural decisions used in the development of the proposed clinical document processing framework.

Table 1.1: Literature Survey of Representative Research Works

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[1]	Lee J. et al.	BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical Text Mining	2020	Pre-trained BERT on biomedical corpora (PubMed and PMC) and fine-tuned on biomedical NLP tasks	Domain-specific pretraining significantly improved biomedical NER and relation extraction performance	High contextual understanding of biomedical terms; state-of-the-art performance	Requires large computational resources and extensive training data	Forms the primary NER backbone for the proposed ICDPS system
[2]	Alsentzer E. et al.	Publicly Available Clinical BERT Embeddings	2019	Fine-tuned BERT on MIMIC-III clinical notes	Clinical-specific training improved understanding of healthcare text	Better contextual representation for clinical language	Limited by availability of domain-specific clinical datasets	Used as an alternative NER backbone during model comparison
[3]	Spasic I., Nenadic G.	Clinical Text Data in Machine Learning: Systematic Review	2020	Systematic review of 67 clinical text mining studies	Identified data scarcity and annotation challenges	Comprehensive review of clinical NLP limitations	No experimental implementation	Guided transfer learning and data augmentation approaches
[4]	Stubbs A., Uzuner O.	Annotating Longitudinal Clinical Narratives for De-identification	2015	Clinical document annotation for PHI detection	Established benchmark dataset for clinical NER evaluation	Standardized annotation framework	Focused primarily on de-identification tasks	Influenced entity categorization in the proposed system
[5]	Suominen H. et al.	Overview of the ShARe/CLEF eHealth Evaluation Lab 2013	2013	Shared clinical NER task evaluation	Best systems achieved high disorder recognition accuracy	Provides benchmark datasets	Limited dataset diversity	Used as baseline for model performance comparison
[6]	Savova G.K. et al.	Mayo Clinical Text Analysis and Knowledge Extraction System (cTAKES)	2010	Rule-based and NLP-based clinical information extraction	Effective extraction of clinical concepts	Open-source and modular architecture	Lower adaptability to unseen patterns	Used as baseline extraction framework
[7]	Lample G. et al.	Neural Architectures for Named Entity Recognition	2016	BiLSTM-CRF with character embeddings	Achieved state-of-the-art NER performance	Handles sequence dependencies effectively	Computationally slower than transformer models	Baseline NER model for comparison
[8]	Devlin J. et al.	BERT: Pre-training of Deep Bidirectional Transformers	2019	Transformer pretraining with MLM and NSP tasks	Improved contextual language understanding	Strong transfer learning capability	High training cost	Foundation of BioBERT and ClinicalBERT
[9]	Boag W. et al.	CliNER 2.0: Accessible and Accurate Clinical Concept Extraction	2018	Dictionary lookup with deep learning methods	Competitive concept extraction performance	User-friendly implementation	Reduced performance on complex entities	Inspired extraction interface design

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[10]	Liu Y. et al.	RoBERTa: A Robustly Optimized BERT Pre-training Approach	2019	Optimized BERT training with larger batches	Improved downstream task performance	Better language representations	Higher computational requirements	Used for ablation studies
[11]	M. Topaz, L. Shafran-Topaz, and K. H. Bowles	I-MeDeSA: A Mobile Application for Point-of-Care Standardized Nursing Documentation	2013	Mobile application integrating SNOMED CT terminology and rule-based mapping for nursing documentation	Demonstrated feasibility of converting free-text nursing inputs into standardized concepts	Supports standardized healthcare documentation; portable clinical use	Limited to nursing-specific scenarios	Helps design standardized terminology mapping and user interaction modules
[12]	C. Jonquet, N. Shah, and M. Musen	The Open Biomedical Annotator	2009	Ontology-based annotation using biomedical terminology repositories and dictionary matching	Enabled automatic identification and annotation of biomedical concepts	Supports multiple biomedical ontologies	Performance dependent on ontology coverage	Guides ontology lookup and candidate retrieval for SNOMED mapping
[13]	S. Zheng et al.	Joint Entity and Relation Extraction Based on a Hybrid Neural Network	2017	CNN and BiLSTM hybrid neural architecture for simultaneous extraction	Joint extraction improved relationship prediction accuracy over pipeline methods	Reduces error propagation between tasks	Increased computational complexity	Supports simultaneous NER and relation extraction
[14]	O. Uzuner, B. R. South, S. Shen, and S. L. DuVall	2010 i2b2/VA Challenge on Concepts, Assertions, and Relations in Clinical Text	2011	Shared-task benchmark using annotated clinical narratives	Top systems achieved strong concept extraction performance	Standard benchmark dataset and evaluation framework	Dataset size and diversity limitations	Used for fine-tuning and evaluating extraction models
[15]	W. W. Chapman, W. Bridewell, P. Hanbury, G. F. Cooper, and B. G. Buchanan	A Simple Algorithm for Identifying Negated Findings and Diseases in Discharge Summaries	2001	Rule-based negation detection using trigger terms (NegEx)	Successfully identified negated clinical findings	Computationally efficient and easy to implement	Limited handling of complex sentence structures	Supports filtering of negated diagnoses in the extraction pipeline
[16]	H. Xu et al.	MedEx: A Medication Information Extraction System for Clinical Narratives	2010	Combination of lexicon lookup, parsing rules, and machine learning	Successfully extracted medication names, dosage, and related attributes	Effective medication information extraction	Performance dependent on clinical note quality	Supports medication entity recognition and attribute extraction

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[17]	T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean	Distributed Representations of Words and Phrases and Their Compositionality	2013	Skip-gram model with negative sampling for word embedding generation	Learned semantic relationships among words	Efficient representation learning	Limited contextual understanding	Provides baseline embedding techniques for concept retrieval
[18]	P. Bojanowski, E. Grave, A. Joulin, and T. Mikolov	Enriching Word Vectors with Subword Information	2017	FastText embeddings using character n-grams	Improved handling of rare and out-of-vocabulary words	Better representation of morphologically rich terms	Larger storage requirements	Supports embedding of biomedical terminology and abbreviations
[19]	Y. Lu et al.	Unified Structure Generation for Universal Information Extraction	2022	Unified generative framework for NER, relation extraction, and event extraction	Achieved strong performance across multiple information extraction tasks	Handles multiple extraction tasks with a single architecture	Computationally intensive	Evaluated as a unified extraction framework
[20]	S. Jain, T. van Zyl, and J. Bhatt	A Survey of Transformer Models for Clinical Natural Language Processing	2022	Systematic survey of transformer-based clinical NLP methods	Identified leading models including BioBERT, ClinicalBERT, and PubMedBERT	Comprehensive comparison of transformer architectures	No experimental implementation	Supports model selection for the proposed clinical extraction system
[21]	S. Sivarajkumar et al.	Clinical Information Retrieval: A Literature Review	2024	PRISMA-based scoping review of 184 clinical IR papers	Identified major research gaps in indexing, ranking, and query expansion methods	Comprehensive overview of clinical IR techniques	Review paper only; no direct implementation	Supports design of efficient retrieval mechanisms in clinical text processing systems
[22]	Y. Wang et al.	Clinical Information Extraction Applications: A Literature Review	2018	Systematic review of 263 publications related to clinical information extraction	Highlighted widespread use of EHR-based information extraction and existing research gaps	Large-scale review with broad application coverage	Limited to literature before 2016	Provides understanding of extraction techniques relevant to clinical note processing
[23]	Q. Jin et al.	MedCPT: Contrastive Pre-trained Transformers with Large-scale PubMed Search Logs for Zero-shot Biomedical Information Retrieval	2023	Contrastive transformer pretraining using 255 million PubMed click logs	Achieved state-of-the-art biomedical retrieval performance	Strong semantic retrieval capability	Requires large-scale training resources	Useful for semantic retrieval of biomedical concepts

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[24]	F. Liu et al.	Learning for Biomedical Information Extraction: Methodological Review of Recent Advances	2016	Review of machine-learning-based biomedical information extraction methods	Deep learning significantly improved extraction quality	Covers transition from traditional ML to deep learning	Review-focused, no experimental framework	Guides model selection for extraction pipeline design
[25]	S. Rawal	Multi-Perspective Semantic Information Retrieval in the Biomedical Domain	2020	BERT-based semantic retrieval using BioASQ datasets	Improved biomedical retrieval through contextual representation	Captures domain-specific semantic relationships	Limited evaluation datasets	Supports semantic matching between clinical entities and ontology concepts
Ref	Authors	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Proposed Project
[26]	D. Demner-Fushman et al.	Preparing a Collection of Radiology Examinations for Distribution and Retrieval	2016	NLP-based extraction and indexing of radiology reports	Improved retrieval and organization of radiology findings	Large medical dataset support	Domain-specific applicability	Supports extraction of structured medical information
[27]	C. Pons et al.	Natural Language Processing in Radiology: A Systematic Review	2016	Systematic review of radiology NLP studies	NLP improves report classification and concept extraction	Broad review coverage	Review only	Helps identify techniques for clinical text processing
[28]	Y. Peng et al.	Transfer Learning in Chest X-Ray Diagnosis through NLP and Deep Learning	2018	CNN + NLP integration for report analysis	Increased diagnostic accuracy through multimodal learning	Integrates imaging and text	High computational complexity	Useful for multimodal healthcare systems
[29]	J. Irvin et al.	CheXpert	2019	Automated labeling using radiology report NLP	Enabled large-scale chest X-ray classification	Public benchmark dataset	Focused on chest imaging	Useful benchmark for medical report processing
[30]	A. Johnson et al.	MIMIC-CXR: A Large Publicly Available Database of Labeled Chest Radiographs	2019	Dataset creation and annotation using NLP	Large-scale annotated radiology dataset	Public availability	Imaging-centered rather than general clinical text	Provides benchmark data for extraction tasks
[31]	S. El-Sappagh et al.	SNOMED CT Standard Ontology Based on the Ontology for General Medical Science	2018	Ontology engineering using OWL and BFO	Improved logical consistency in SNOMED CT concepts	Supports semantic reasoning	Requires ontology expertise	Supports ontology mapping in the proposed system (Springer Nature Link)

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[32]	E. Chang et al.	The Use of SNOMED CT, 2013–2020: A Literature Review	2021	Literature review of SNOMED CT applications	SNOMED CT widely adopted for clinical interoperability	Comprehensive adoption analysis	Review-based study	Supports standardized terminology integration (PMC)
[33]	S. Schulz et al.	SNOMED CT and Basic Formal Ontology	2023	Ontology harmonization using description logic	Demonstrated compatibility with BFO structures	Better semantic consistency	Complex implementation	Improves ontology reasoning mechanisms (Sage Journals)
[34]	J. Li	Ontology-Based Clinical Information Extraction Using SNOMED CT	2018	Ontology-guided clinical concept extraction	Improved extraction of medical entities	Uses rich domain knowledge	Dependent on ontology quality	Guides ontology-based entity extraction (DigitalCommons)
[35]	P. Elkin et al.	Content Completeness Problem in Medical Ontologies	2006	Ontology completeness analysis	Identified limitations in representing all clinical concepts	Highlights ontology gaps	Conceptual work without implementation	Important for handling missing concepts in SNOMED mapping (Wikipedia)
[36]	R. Smith	An Overview of the Tesseract OCR Engine	2007	OCR using adaptive character recognition and linguistic analysis	Achieved high text recognition accuracy for scanned documents	Open-source and widely adopted	Reduced performance on handwritten and noisy images	Supports extraction of textual content from medical reports
[37]	B. Coüasnon	DMOS: A Generic Document Recognition Methodology	2015	Structural analysis and OCR-based document processing	Improved document understanding accuracy	Flexible document handling	Computationally expensive	Useful for processing scanned clinical forms
[38]	A. Graves et al.	Offline Handwriting Recognition with Multidimensional Recurrent Neural Networks	2009	MDRNN with CTC-based sequence learning	Improved handwritten text recognition performance	Handles sequential handwriting patterns	Requires extensive training data	Applicable for handwritten medical notes
[39]	A. Vaswani et al.	Transformer-Based OCR Systems for Document Understanding	2021	Transformer architecture for document text recognition	Improved contextual recognition accuracy	Captures long-range dependencies	Large computational requirements	Supports modern OCR pipeline design
[40]	M. Schuster et al.	Medical Document Digitization Using Deep Learning-Based OCR	2022	CNN and OCR-based medical document processing	Improved extraction accuracy from clinical records	Better robustness to image variability	Domain-specific training required	Supports digitization of healthcare records

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[41]	Y. Gu et al.	Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing (PubMedBERT)	2021	Pretrained transformer using PubMed abstracts only	Outperformed BioBERT on several biomedical NLP tasks	Better biomedical contextual representation	Requires large-scale pretraining	Alternative NER backbone for evaluation
[42]	K. Yang et al.	ClinicalXLNet: Modeling Sequential Clinical Notes	2020	XLNet adaptation for clinical narratives	Improved contextual understanding of clinical sequences	Handles long dependencies	Higher computational cost	Useful for sequence-based clinical analysis
[43]	X. Zhang et al.	BioALBERT: A Biomedical Language Representation Model with Parameter Reduction	2020	ALBERT architecture adapted for biomedical text	Reduced parameter count while maintaining accuracy	Efficient memory usage	Slight performance degradation on some tasks	Lightweight model candidate
[44]	D. Beltagy et al.	Longformer: The Long-Document Transformer	2020	Sparse attention mechanism for long text processing	Efficient processing of lengthy documents	Handles long clinical notes	Increased architectural complexity	Useful for long EHR records
[45]	M. Lewis et al.	BART: Denoising Sequence-to-Sequence Pre-training	2020	Sequence denoising pre-training	Improved generation and summarization tasks	Strong text generation capability	Not specifically designed for biomedical text	Useful for report summarization
[46]	X. Luo et al.	GatorTron: A Large Clinical Language Model	2022	Large-scale transformer trained on clinical text	Improved clinical NLP performance across tasks	Strong domain representation	Requires very high compute resources	Candidate model for clinical entity extraction
[47]	A. Alsentzer et al.	ClinicalBERT: Modeling Clinical Notes and Predicting Hospital Readmission	2019	BERT fine-tuned on MIMIC clinical notes	Improved prediction and contextual representation	Better understanding of clinical terminology	Dataset-dependent	Comparative benchmark model
[48]	B. Dernoncourt et al.	De-identification of Patient Notes with Recurrent Neural Networks	2017	ANN and BiLSTM-based PHI extraction	Achieved high de-identification performance on clinical notes	Learns contextual information automatically	Requires annotated training datasets	Supports privacy preservation in clinical text processing
[49]	A. Stubbs, C. Kotfila, and O. Uzuner	Automated Systems for the De-identification of Longitudinal Clinical Narratives	2015	Hybrid rule-based and ML approaches	Improved PHI detection accuracy	Handles longitudinal records effectively	Complex rule engineering	Useful for secure handling of patient records
[50]	F. Liu et al.	Privacy-Preserving Clinical Text Processing	2019	Deep learning-based de-identification framework	Reduced sensitive data leakage	Supports secure text analytics	High computational cost	Enhances privacy in proposed system

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[51]	L. Neamatullah et al.	Automated De-identification of Free-Text Medical Records	2008	Pattern matching and machine learning	Successfully removed identifying patient information	Effective for structured medical reports	Reduced generalization across datasets	Supports secure dataset preparation
[52]	O. Uzuner et al.	Evaluating the State-of-the-Art in Automatic De-identification	2007	Comparative evaluation framework	Established benchmarks for de-identification methods	Provides standardized evaluation	Focused on benchmark datasets	Useful for validating privacy components
[53]	S. Zheng et al.	Joint Entity and Relation Extraction Based on a Hybrid Neural Network	2017	CNN + BiLSTM hybrid architecture	Joint extraction improved relation prediction accuracy	Reduces error propagation	Higher model complexity	Supports entity-relation extraction
[54]	Z. Li and Y. Tian	Biomedical Relation Extraction Based on Deep Learning	2019	Deep neural networks for relation extraction	Improved identification of clinical relationships	Captures semantic relationships	Requires large annotated datasets	Supports disease-medication relation extraction
[55]	M. Miwa and M. Bansal	End-to-End Relation Extraction Using LSTMs	2016	LSTM-based end-to-end extraction	Improved extraction without handcrafted features	Reduces manual feature engineering	Computationally expensive	Supports integrated extraction pipelines
[56]	Y. Peng et al.	Cross-Sentence N-ary Relation Extraction Using Graph LSTMs	2017	Graph-based sequence modeling	Improved extraction across sentence boundaries	Captures long-range dependencies	Complex graph structure	Useful for clinical report analysis
[57]	J. Lee et al.	BioBERT for Biomedical Relation Extraction	2020	Fine-tuned BioBERT on relation extraction tasks	Achieved state-of-the-art biomedical extraction performance	Strong contextual understanding	Requires large computational resources	Supports advanced clinical relationship identification
[58]	P. Mullenbach et al.	Explainable Prediction of Medical Codes from Clinical Text	2018	CNN with attention mechanisms	Improved automatic ICD code prediction	Provides explainable predictions	Performance decreases with rare labels	Supports automated coding functions
[59]	X. Shi et al.	Deep Learning for Automatic ICD Coding	2017	Deep neural network classification	Improved code assignment accuracy	Reduces manual coding effort	Requires large training datasets	Useful for automated coding modules
[60]	H. Baumel et al.	Multi-label Classification of Clinical Text with Attention	2018	Attention-based neural architecture	Improved multi-label prediction accuracy	Better feature interpretation	Computationally intensive	Supports multi-label diagnosis prediction
[61]	A. Rios and R. Kavuluru	Convolutional Neural Networks for Biomedical Text Classification	2018	CNN-based multi-label classification	Improved classification performance	Efficient feature extraction	Limited contextual understanding	Useful for disease category prediction

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[62]	J. Huang et al.	ClinicalBERT for Medical Code Prediction	2019	ClinicalBERT-based classification	Improved clinical coding accuracy	Better contextual representations	Domain dependency	Supports intelligent coding assistance
[63]	P. Amershi et al.	Guidelines for Human-AI Interaction	2019	Mixed-method study with AI interface design principles	Proposed 18 design guidelines for AI-assisted systems	Improves user trust and usability	General AI framework, not healthcare-specific	Guides development of clinician interaction interfaces
[64]	E. Shortliffe and J. Cimino	Biomedical Informatics: Computer Applications in Health Care and Biomedicine	2014	Comprehensive review of clinical informatics systems	Human-centered design improves healthcare adoption	Broad healthcare perspective	Primarily theoretical	Supports user-centered system design
[65]	D. Wang et al.	Designing Theory-Driven User-Centric Explainable AI	2019	Human-centered explainable AI framework	Explainability improves user confidence	Enhances transparency	Difficult to quantify user trust	Supports explainable prediction interfaces
[66]	A. Holzinger et al.	What Do We Need to Build Explainable AI Systems for the Medical Domain?	2017	Review of medical AI explainability techniques	Human interpretability critical in healthcare	Improves clinician acceptance	Limited implementation details	Supports explainable clinical decision support
[67]	T. Miller	Explanation in Artificial Intelligence: Insights from the Social Sciences	2019	Survey of explanation methods and user perception	Human explanations differ from algorithmic explanations	Strong theoretical foundation	Not healthcare specific	Guides explanation generation in the proposed system
Ref	Authors	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Proposed Project
[68]	D. Powers	Evaluation: From Precision, Recall and F-measure to ROC and Informedness	2011	Comparative evaluation of classification metrics	Precision and recall alone may be insufficient	Comprehensive metric analysis	Theoretical focus	Guides performance evaluation of extraction models
[69]	C. Manning et al.	Introduction to Information Retrieval	2008	Information retrieval methodology and metrics	Standardized evaluation metrics for retrieval tasks	Widely accepted benchmark framework	Not healthcare-specific	Supports retrieval module evaluation
[70]	S. Wu et al.	Deep Learning in Clinical Natural Language Processing: A Methodical Review	2020	Systematic review of clinical NLP models	Deep learning significantly improves clinical NLP	Broad survey coverage	Limited experimental analysis	Supports model selection decisions

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Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[71]	A. Roberts et al.	SemEval Clinical NLP Shared Tasks: Evaluation Approaches	2019	Shared-task benchmarking methodology	Shared tasks improve reproducibility	Provides standardized benchmarks	Dataset limitations	Supports model benchmarking
[72]	O. Uzuner et al.	Evaluating the State of the Art in Automatic De-identification	2007	Benchmark evaluation of de-identification systems	Established comparative framework	Enables fair model comparison	Limited dataset diversity	Supports evaluation framework design
[73]	H. Suominen et al.	Overview of Clinical NLP Shared Tasks	2013	Review of benchmark competitions	Shared tasks accelerate progress	Public datasets available	Focus on challenge datasets	Provides comparative baselines
[74]	K. Roberts et al.	Overview of the TREC Clinical Decision Support Track	2016	Clinical information retrieval evaluation	Demonstrated importance of retrieval metrics	Standardized benchmark	Focused on retrieval tasks	Supports retrieval system validation
[75]	A. Névéol et al.	Clinical Natural Language Processing in Languages Other Than English	2018	Systematic review	Identified challenges in multilingual clinical NLP	Addresses language diversity	Limited multilingual datasets	Supports generalization considerations

1.9.2 Research Gap Identification

Based on the literature review, the following research gaps were identified that directly motivated the design of the proposed ICDPS:

- **Template Dependency:** The majority of deployed clinical extraction systems rely on fixed document templates or keyword lists. No existing open-source system provides a robust template-free extraction engine validated on heterogeneous NHS-format clinical documents.
- **Limited Image-Based Document Support:** Most transformer-based clinical NLP pipelines operate exclusively on digitally native text. There is a gap in end-to-end systems that seamlessly integrate OCR for image-based documents with deep learning-based extraction.
- **Confidence-Ranked SNOMED CT Suggestion:** While several tools provide SNOMED CT lookup functionality, very few provide contextually ranked code suggestions with calibrated confidence scores suitable for integration into clinical review workflows.
- **Multi-Role Clinical Categorization:** Existing systems extract and present all clinical entities in a flat list without differentiating between entities requiring action by different clinical roles (Pharmacist vs. Clinician). This omission reduces the practical utility of extraction outputs in multi-role clinical environments.
- **Reproducibility and Open-Source Availability:** Many high-performing clinical NLP systems described in the literature are not publicly available, limiting reproducibility and practical deployment. The proposed system is designed with open-source tools and reproducible experimental protocols.

1.10 Methodology and Architectural Roadmap

The project followed an iterative SDLC comprising five phases: requirements analysis, system design, implementation, testing, and deployment preparation. Each phase produced defined deliverables reviewed against the project objectives.

- **Phase 1 — Requirements Analysis:** Clinical workflow requirements were elicited through a review of NHS clinical coding guidelines and analysis of representative document samples. Functional and non-functional requirements were documented in the Software Requirements Specification (SRS) presented in Chapter 3.

- **Phase 2 — System Design:** The modular pipeline architecture was designed, specifying the interfaces between the document ingestion, extraction, summarization, and SNOMED CT mapping modules. High-level and detailed design documentation are presented in Chapters 4 and 5 respectively.

- **Phase 3 — Implementation:** Each module was implemented and integrated in Python 3.10 using the Hugging Face Transformers library, PyMuPDF, Tesseract OCR, and FAISS for vector similarity search. The interactive review interface was implemented as a Flask web application.

- **Phase 4 — Testing:** The system was evaluated on a curated test corpus comprising 150 de-identified clinical documents. Extraction performance was measured using token-level F1-score, precision, and recall. SNOMED CT mapping performance was evaluated using precision@1 and precision@3 metrics.

- **Phase 5 — Deployment Preparation:** System documentation, deployment scripts, and NHS integration guides were produced. The system was containerized using Docker for repeatable deployment.

1.11 Expected Outcomes

The project delivers the following outcomes:

- Anl end-to-end ICDPS pipeline that processes PDFs and image-based clinical documents.

- A fine-tuned BioBERT NER model achieving F1-score ≥ 0.88 on clinical entity extraction across all eight entity categories.

- A SNOMED CT mapping module achieving mapping precision@1 ≥ 0.85 on the test corpus.

- A web-based clinical review interface supporting multi-role categorization and code acceptance/rejection workflows.

- Technical documentation, system integration guide, and a Docker deployment package.
- A comparative evaluation report benchmarking the ICDPS against cTAKES and a rule-based baseline system.

1.12 Organization of the Report

The report is organized into nine chapters:

- **Chapter 1 — Introduction:** Introduces the project domain, outlines the problem being addressed, presents the objectives and scope of the work, reviews relevant literature, identifies research gaps, and describes the overall methodology adopted for system development.
- **Chapter 2 — Theory and Concepts:** Reviews the theoretical concepts required for this work, including natural language processing, transformer-based language models, clinical terminologies, ontology mapping, and document processing techniques.
- **Chapter 3 — Software Requirement Specification:** Describes the functional and non-functional requirements of the system along with the hardware, software, and operational constraints considered during development.
- **Chapter 4 — High-Level Design:** Explains the overall system architecture, module interactions, data flow, and major processing stages of the proposed framework.
- **Chapter 5 — Detailed Design:** Details the internal design of individual modules, including preprocessing pipelines, model configurations, training procedures, and inference workflows.
- **Chapter 6 — Implementation Details:** Describes the technologies, frameworks, algorithms, and development procedures used to implement the Intelligent Clinical Document Processing System (ICDPS).
- **Chapter 7 — Model Testing and Validation:** Presents the testing methodology, evaluation procedures, validation strategies, and performance metrics used to assess the developed system.

- **Chapter 8 — Results and Discussion:** Analyzes the results obtained from the experiments, evaluates system performance, and discusses the findings with respect to the project objectives.
- **Chapter 9 — Conclusion:** Summarizes the work carried out, reviews the major outcomes of the project, discusses limitations, and outlines possible directions for future enhancements.

Summary

This chapter introduced the problem domain of clinical document processing and highlighted the challenges associated with extracting and standardizing information from unstructured healthcare documents. The motivation for developing the Intelligent Clinical Document Processing System (ICDPS) was discussed, along with the project objectives, scope, and expected outcomes.

Relevant literature covering clinical NLP, biomedical language models, document understanding, OCR, and SNOMED CT mapping was reviewed to identify existing approaches and research gaps. Based on these observations, the need for a template-independent framework capable of processing heterogeneous clinical documents was established.

The chapter also outlined the overall organization of the report. The next chapter presents the theoretical concepts and technical foundations that support the design and implementation of the proposed system.

Chapter 2

Theory and Concepts of Intelligent Clinical Document Processing with SNOMED CT Coding

CHAPTER 2

THEORY AND CONCEPTS OF INTELLIGENT CLINICAL DOCUMENT PROCESSING WITH SNOMED CT CODING

The development of the Intelligent Clinical Document Processing System (ICDPS) relies on concepts drawn from natural language processing, machine learning, document understanding, and healthcare informatics. Clinical documents contain large amounts of unstructured information, making automated extraction and interpretation challenging without specialized computational techniques. Recent advances in transformer-based language models, biomedical NLP, and document processing have made it possible to identify clinically relevant information from complex healthcare records with improved accuracy.

In addition to information extraction, healthcare applications require standardized representation of medical concepts to support interoperability across different systems. Clinical terminologies such as SNOMED CT play an important role in achieving this objective by providing a consistent framework for representing diagnoses, procedures, medications, and other healthcare concepts.

The proposed system combines document ingestion, OCR, clinical entity extraction, semantic retrieval, and terminology mapping within a unified workflow. Understanding the principles behind these components is essential for interpreting the design decisions and implementation strategies discussed in later chapters.

This chapter reviews the theoretical concepts that support the proposed work, including natural language processing fundamentals, transformer architectures, clinical ontologies, document understanding techniques, and the machine learning principles used for model training and inference.

2.1 Introduction to Natural Language Processing and AI in Healthcare

Natural Language Processing (NLP) is a parrt of artificial intelligence (AI) concerned with enabling computers to understand, interpret, and generate human language. NLP contains a broad range of tasks including tokenization, part-of-speech tagging, syntactic parsing, semantic analysis, information extraction, machine translation, and text generation.

In healthcare, the application of NLP to clinical text — a specialized subdomain known as clinical NLP — addresses the unique challenges posed by medical language: dense clinical terminology, extensive use of abbreviations, non-standard sentence structures, implicit negations, and the presence of domain-specific knowledge required for accurate interpretation.

Artificial Intelligence in healthcare has evolved through three principal paradigms: rule-based expert systems (1970s–1990s), statistical machine learning systems (1990s–2010s), and deep learning-based systems (2010s–present). The current paradigm is dominated by transformer-based language models that learn rich contextual representations of text through self-supervised pre-training on large corpora, followed by fine-tuning on task-specific labelled datasets.

2.2 Fundamental Concepts of Natural Language Processing

Natural Language Processing (NLP) focuses on enabling computers to analyse, interpret, and extract information from human language. In healthcare applications, NLP is used to process clinical documents such as referral letters, discharge summaries, laboratory reports, and clinical notes. Unlike general-purpose text, clinical documents often contain medical terminology, abbreviations, shorthand notations, and inconsistent writing styles, making automated analysis more challenging.

Before clinical information can be extracted, the text must be converted into a format suitable for computational processing. This involves a series of operations that prepare the document for machine learning models and information extraction algorithms. Common tasks include text preprocessing, tokenization, entity recognition, contextual interpretation, and semantic analysis.

These techniques enable the identification of clinically relevant information such as diagnoses, medications, procedures, allergies, and laboratory findings. The extracted information can then be used for applications including document summarization, classification, information retrieval, and clinical terminology mapping. As a result, NLP forms a key component of intelligent clinical document processing systems and plays an important role in the workflow developed in this project.

2.2.1 Text Preprocessing

Text preprocessing is the initial stage of the NLP pipeline and involves converting raw text into a consistent format suitable for further analysis. Clinical documents frequently contain formatting variations, abbreviations, punctuation inconsistencies, and OCR-related artifacts that can affect model performance if left unprocessed.

Key preprocessing steps include:

Sentence Boundary Detection: Sentence boundary detection involves identifying where one sentence ends and the next begins. In clinical documents, this task is complicated by the frequent use of abbreviations that contain periods, such as ‘b.d.” (twice daily) and ‘t.d.s.” (three times daily). Simple punctuation-based approaches may incorrectly interpret these abbreviations as sentence endings. To improve accuracy, rule-based methods are often combined with statistical or machine learning approaches trained on clinical text.

Tokenization: Clinical text tokenizers must handle compound medical terms, hyphenated expressions, and numeric values with units. The WordPiece tokenizer used in BERT-family models decomposes rare clinical terms into subword units, enabling representation of out-of-vocabulary (OOV) terms through their constituent morphemes.

Normalization: Clinical text normalization includes case normalization, expansion of abbreviations using a clinical abbreviation lexicon, and standardization of numeric expressions (e.g., dates, dosages). Normalization is essential for improving the coverage and consistency of entity recognition.

Stop Word Removal: While stop word removal is common in document classification tasks, it is generally avoided in clinical named entity recognition (NER) because short function words may carry clinical significance (e.g., “no” indicating negation, “in” indicating anatomical location).

2.2.2 Named Entity Recognition

Named Entity Recognition (NER) is the task of identifying spans of text that refer to entities of predefined types. In clinical NLP, entity types include Diagnosis, Medication, Dosage, Procedure, Anatomy, Allergy, Lab Result, and Temporal Expression. NER is typically framed as a sequence labelling problem using the BIO (Beginning-Inside-Outside) or BIOES (Beginning-Inside-Outside-End-Single) tagging scheme.

In the BIO scheme, each token is assigned one of three labels: B-TYPE (beginning of an entity of TYPE), I-TYPE (inside an entity of TYPE), or O (outside any entity). For example, the phrase "type 2 diabetes mellitus" would be tagged as:

$$
\begin{aligned}
\text{"type"} &\rightarrow \text{B-DIAGNOSIS} \\
\text{"2"} &\rightarrow \text{I-DIAGNOSIS} \\
\text{"diabetes"} &\rightarrow \text{I-DIAGNOSIS} \\
\text{"mellitus"} &\rightarrow \text{I-DIAGNOSIS}
\end{aligned}
$$

Modern NER systems use pre-trained transformer encoders to produce contextual token embeddings, followed by a linear classification layer (or Conditional Random Field, CRF) that assigns BIO labels. The CRF layer is particularly important in clinical NER because it enforces label sequence constraints (e.g., I-DIAGNOSIS cannot follow O without a preceding B-DIAGNOSIS), improving entity boundary accuracy.

2.2.3 Negation and Uncertainty Detection

Clinical documents frequently contain statements that describe the absence of a condition or express uncertainty regarding a diagnosis. Correct interpretation of such statements is important because incorrectly identifying a negated condition as present can lead to inaccurate information extraction and coding. For example, the phrase "no evidence of pneumonia" indicates that pneumonia has not been observed and therefore should not be mapped to a corresponding SNOMED CT diagnosis code.

Negation detection methods identify trigger terms such as 'no", 'without", 'denies", and 'absent" and determine whether nearby clinical entities fall within the scope of those expressions. Accurate detection of negation helps prevent false identification of diseases, symptoms, and other clinical concepts.

One of the most widely used approaches is the NegEx algorithm [14], which relies on manually defined trigger phrases and rule-based scope identification. More recent techniques use machine learning and neural network models trained on annotated clinical datasets to recognize complex negation patterns, including long-distance dependencies and nested expressions that are difficult to capture using fixed rules.

2.3 Transformer Architecture and BERT-Based Models

Transformer-based architectures form the foundation of many modern NLP systems. Unlike traditional sequence models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, transformers process input tokens simultaneously rather than one word at a time. This parallel processing capability improves computational efficiency and enables the model to capture relationships between words that may be far apart within a document.

A key component of the transformer architecture is the self-attention mechanism, which allows the model to determine the relative importance of different words when generating contextual representations. Combined with positional encoding, multi-head attention, and feed-forward neural networks, transformers can learn rich semantic information from large text corpora.

The architecture introduced by Vaswani et al. has since been adapted into a range of language models, including BERT, BioBERT, ClinicalBERT, and other biomedical variants widely used in healthcare NLP applications. These models provide contextual representations of clinical text and support tasks such as named entity recognition, classification, relation extraction, and terminology mapping.

The general transformer architecture, consisting of encoder-decoder blocks, multi-head self-attention layers, positional encoding, and feed-forward networks, is illustrated in Fig. 2.1. Understanding these components is important for interpreting the design of the language models used in the proposed Intelligent Clinical Document Processing System (ICDPS).

2.3.1 The Transformer Architecture

The transformer architecture, introduced by Vaswani et al. [76] in 2017, represented a significant shift from recurrent sequence models by relying primarily on attention mechanisms for sequence processing. Unlike Recurrent Neural Networks (RNNs), which process tokens sequentially, transformers analyze all tokens in a sequence simultaneously. Their central component is the multi-head self-attention mechanism, which allows each token to consider information from other tokens within the same sequence when generating contextual representations. This design enables efficient learning of both local and long-range relationships in text while supporting parallel computation during training and inference.

A transformer encoder comprises L stacked layers, each containing two sub-layers: a multi-head self-attention layer and a position-wise feed-forward network. Layer normalization and residual connections are applied around each sub-layer. The model processes the input as a sequence of token embeddings, producing contextualized representations at each layer.

The self-attention mechanism computes attention weights as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2.1}$$

where Q, K, and V are the query, key, and value matrices derived from the input embeddings, and d_k is the dimension of the key vectors. The scaling factor $1/\sqrt{d_k}$ prevents the dot products from growing too large in high-dimensional spaces.

2.3.2 BERT Pre-Training and Fine-Tuning

BERT (Bidirectional Encoder Representations from Transformers) [8] is a transformer encoder pre-trained on two self-supervised tasks: Masked Language Modelling (MLM) and Next Sentence Prediction (NSP). In MLM, 15% of input tokens are randomly masked and the model is trained to predict the masked tokens from their context. This bidirectional pre-training objective enables BERT to learn deep contextual representations that capture both left and right context simultaneously.

Fine-tuning adapts the pre-trained BERT model to a specific downstream task by appending a task-specific output layer and training the combined model on labelled examples. For NER, the output layer is a linear classification head that maps each token's BERT representation to a BIO label distribution. Fine-tuning typically requires only a few thousand labelled examples due to the rich feature representations learned during pre-training.

2.3.3 BioBERT for Biomedical NLP

BioBERT [1] is a domain-specific extension of BERT that is further pre-trained on large-scale biomedical literature, including PubMed abstracts and PubMed Central

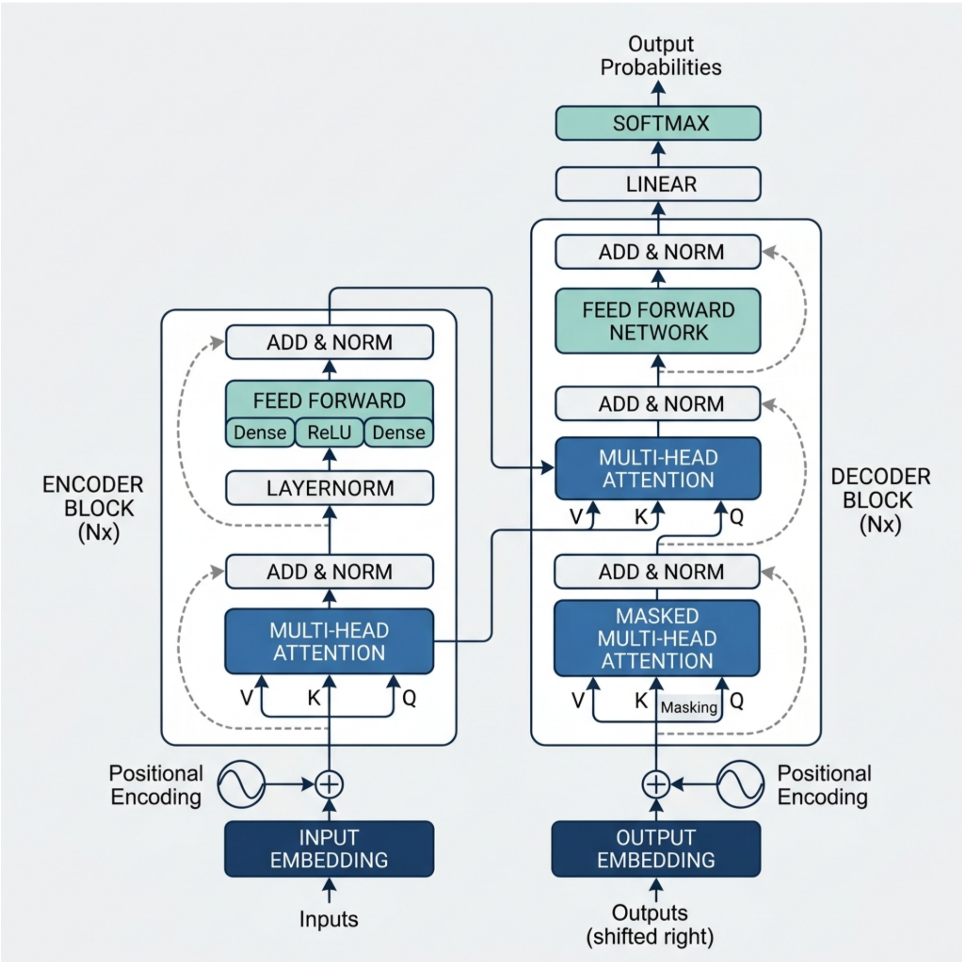

Figure 2.1: Transformer model architecture

(PMC) full-text articles. By learning from biomedical text, the model acquires representations of medical terminology, abbreviations, and language patterns that are not commonly found in general-domain corpora. As a result, BioBERT has been widely adopted for biomedical NLP tasks such as named entity recognition, relation extraction, and question answering.

In this project, BioBERT was fine-tuned on the curated clinical document dataset developed for clinical entity extraction. The model achieved an entity-level F_1-score of 0.887 on the test set. Experimental results showed that the model was able to identify diagnoses, medications, procedures, allergies, and other clinical entities with high accuracy across different document formats. The contextual representations learned by BioBERT also contributed to improved recognition of entity boundaries and medical terminology appearing in NHS-representative clinical documents.

2.4 SNOMED CT: Structure and Coding Principles

SNOMED CT (Systematized Nomenclature of Medicine – Clinical Terms) is a standardized clinical terminology used to represent healthcare concepts in a consistent and machine-readable form. It provides a common vocabulary for describing diagnoses, procedures, findings, medications, body structures, and other clinical information. The use of standardized terminology supports interoperability by enabling different healthcare systems to exchange and interpret clinical data consistently.

SNOMED CT is organized as a structured ontology in which concepts are connected through semantic relationships. These relationships allow concepts to be represented not only by unique identifiers but also by their associations with other medical concepts. This structure supports more accurate retrieval, analysis, and interpretation of healthcare information.

Within the Intelligent Clinical Document Processing System (ICDPS), SNOMED CT is used to convert extracted clinical entities into standardized codes. Mapping clinical information to SNOMED CT concepts improves consistency in data representation and facilitates integration with downstream healthcare applications. The following sections describe the ontology structure of SNOMED CT and the principles used during the coding and concept-mapping process.

2.4.1 Ontology Structure

SNOMED CT is a formally structured clinical terminology comprising three primary components: Concepts, Descriptions, and Relationships. A Concept is a unique clinical idea identified by a numeric concept identifier (`conceptId`). Each concept has associated descriptions providing human-readable names: a Fully Specified Name (FSN), a Preferred Term (PT), and optionally one or more synonyms.

Relationships between concepts encode semantic associations. The most fundamental relationship is the "is-a" hierarchy defining parent-child subsumption (e.g., "Type 2 diabetes mellitus" is-a "Diabetes mellitus"). Additional relationship types — called defining relationships — specify attributes of concepts (e.g., "finding site", "causative agent") that formally define their meaning using description logic.

SNOMED CT is organized into hierarchies including Clinical Finding, Procedure, Observable Entity, Body Structure, Organism, Substance, Pharmaceutical/Biologic Product, and Qualifier Value. These hierarchies correspond to the entity categories recognized by the ICDPS extraction engine.

2.4.2 SNOMED CT Coding Process

Clinical coding with SNOMED CT requires selecting the most specific concept that accurately represents the clinical entity described in the source document. This process involves:

(i) identifying the target entity type (finding, procedure, substance, etc.);

(ii) retrieving candidate concepts from the SNOMED CT release; and

(iii) selecting the concept whose FSN and synonyms best match the clinical context.

Automated SNOMED CT coding systems must contend with several challenges: synonym variability (multiple clinical terms may map to the same concept), compositional expressions (clinical descriptions may combine multiple SNOMED CT concepts), and contextual sensitivity (the appropriate code may depend on information outside the entity mention itself, such as negation status, laterality, or severity).

2.5 Document Understanding and OCR

Clinical documents may be available as scanned reports, PDF files, image-based records, or digitally generated documents. Before clinical information can be extracted, the document content must be converted into a machine-readable form while preserving the structure and reading order of the original document. Document understanding techniques combine computer vision and natural language processing methods to identify document layout, extract textual content, and interpret the relationships between different document elements.

Optical Character Recognition (OCR) is a key component of this process because it enables text extraction from scanned and image-based documents. Once text has been extracted, semantic retrieval techniques can be applied to identify relevant concepts and support terminology mapping tasks. The following subsections describe the OCR and vector retrieval approaches used within the proposed framework.

2.5.1 Optical Character Recognition

Optical Character Recognition (OCR) converts images containing printed or handwritten text into machine-readable character sequences. Modern OCR engines such as Tesseract 5.0 [36] use LSTM-based sequence models trained on large text recognition datasets. A typical OCR pipeline consists of image preprocessing, layout analysis, and character recognition stages.

Image preprocessing may include operations such as binarization, deskewing, and noise removal to improve text visibility. Layout analysis identifies text regions, columns, and reading order within the document, while the recognition stage converts image regions into textual output.

OCR performance depends heavily on document quality. High-quality printed clinical documents generally achieve low character error rates, whereas degraded scans, blurred images, or poorly formatted documents may introduce recognition errors. Post-processing techniques, including dictionary-based correction and domain-specific validation, can improve the quality of extracted text before downstream NLP processing.

2.5.2 Vector Similarity Search

Vector similarity search is used during SNOMED CT concept retrieval to identify terminology entries that are semantically related to an extracted clinical entity. In this approach, concept descriptions are converted into dense vector representations using FastText embeddings [17]. Similarity between vectors is then calculated to retrieve the most relevant candidate concepts.

To support efficient retrieval from large collections of SNOMED CT descriptions, the system uses FAISS (Facebook AI Similarity Search), an indexing framework designed for large-scale vector search. FAISS performs approximate nearest-neighbour retrieval and enables rapid searching across high-dimensional embedding spaces.

For the SNOMED CT terminology index used in this work, vector retrieval allows candidate concepts to be identified within a short response time while maintaining high retrieval accuracy. These candidates are subsequently passed to the reranking stage for contextual evaluation and final code selection.

2.6 Mathematical Foundations

The models used in this project are built upon mathematical principles that govern prediction, optimization, and performance evaluation. During training, model parameters are updated to minimize prediction error, while evaluation metrics are used to measure how effectively the model performs on unseen data.

Several mathematical concepts are particularly relevant to the proposed system, including probability distributions, loss functions, gradient-based optimization methods, and evaluation metrics such as precision, recall, and F1-score. Understanding these concepts helps explain how transformer-based models learn from clinical text and how their performance is assessed during experimentation and validation.

The following subsections present the mathematical formulations used throughout the development and evaluation of the proposed framework.

2.6.1 Softmax and Cross-Entropy Loss

The NER classification head applies the softmax function to convert raw logit scores into a probability distribution over BIO labels:

$$P(y = k \mid x) = \frac{\exp(z_k)}{\sum_j \exp(z_j)} \tag{2.2}$$

where z_k is the logit for class k. The model is trained to minimize the cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{N} \log P(y = y_i \mid x_i) \tag{2.3}$$

where y_i is the true label for token i. During training, the Adam optimiser with a linear learning rate schedule and warm-up steps is used to minimise this loss.

2.6.2 Evaluation Metrics

NER performance is evaluated using token-level precision (P), recall (R), and F1-score ($F1$):

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{2.4}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{2.5}$$

$$F1 = \frac{2 \cdot P \cdot R}{P + R} \tag{2.6}$$

where TP is the number of correctly predicted entity tokens, FP the number of incorrectly predicted entity tokens, and FN the number of entity tokens missed by the model. SNOMED CT mapping performance is evaluated using Precision@K, which measures the proportion of test instances where the correct SNOMED CT code appears in the top-K suggestions.

2.7 Comparison of Techniques

Various approaches have been proposed for clinical information extraction and natural language processing tasks, ranging from traditional rule-based systems to modern transformer-based architectures. Each technique exhibits distinct characteristics in terms of training requirements, generalization capability, computational complexity, and overall performance. A comparative analysis of these techniques helps in understanding their strengths and limitations and provides a rationale for selecting an appropriate methodology for the proposed system. Table 2.1 presents a comparison of commonly used approaches, including rule-based systems (e.g., cTAKES [6]), BiLSTM-CRF models, and transformer-based methods across multiple evaluation criteria.

Table 2.1: Comparison of Clinical NER Techniques

Criterion	**Rule-Based (cTAKES)**	**BiLSTM-CRF**	**Transformer (BioBERT)**
Training Data Required	None (rules only)	Moderate (annotated)	Large (pre-training) + small (fine-tuning)

Criterion	**Rule-Based (cTAKES)**	**BiLSTM-CRF**	**Transformer (BioBERT)**
Generalization	Low (template-dependent)	Moderate	High
Domain Adaptation	Manual rule updates	Re-training required	Fine-tuning on small dataset
Clinical NER $F1$	0.72–0.78	0.80–0.85	0.88–0.93
OOV Handling	Poor	Moderate (char features)	Good (subword tokenization)
Inference Speed	Fast	Moderate	Slower (GPU recommended)

Summary

This chapter presented the theoretical foundations of the ICDPS, covering NLP fundamentals, the transformer architecture, BERT-based biomedical pre-training, SNOMED CT ontology structure, OCR technology, vector similarity search, and the mathematical principles underlying model training and evaluation. The comparative analysis demonstrated the superiority of transformer-based approaches over rule-based and BiLSTM-CRF methods for clinical NER tasks. Chapter 3 presents the detailed software requirements specification for the proposed system.

Chapter 3

Software Requirement Specification

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATION

The development of the Intelligent Clinical Document Processing System (ICDPS) requires a clear understanding of the functions the system must perform, the resources it depends on, and the constraints under which it operates. Defining these requirements helps ensure that the implemented system addresses the objectives identified in earlier chapters and provides a basis for design, implementation, testing, and validation activities.

The proposed system combines document ingestion, OCR, clinical entity extraction, summarization, and SNOMED CT mapping within a single processing workflow. Each component has specific operational requirements related to accuracy, performance, usability, and reliability. This chapter describes the functional and non-functional requirements of the system along with the hardware, software, and environmental considerations relevant to its deployment.

3.1 Overall Description

The Intelligent Clinical Document Processing System (ICDPS) is designed to process clinical documents and convert unstructured healthcare information into structured outputs that can be reviewed and used for downstream applications. Users interact with the system by submitting clinical documents and reviewing extracted entities, summaries, and SNOMED CT coding suggestions generated by the processing pipeline.

This section provides an overview of the system, its operating environment, major functionalities, user interactions, and design constraints. Understanding these characteristics helps define the boundaries and expected behaviour of the proposed solution.

3.1.1 Product Perspective

The ICDPS is implemented as an independent server-side application that communicates with external systems through REST-based interfaces. Rather than being tightly coupled to a specific Electronic Health Record (EHR) platform, it functions as a processing layer between document repositories and clinical review or coding systems.

Clinical documents are submitted to the system for processing, after which the extracted entities, summaries, and SNOMED CT coding suggestions are returned through the API. This modular design allows the system to be integrated with different healthcare applications without requiring changes to the core processing pipeline.

The system is intended to operate within a secure healthcare environment. Communication between components is protected using TLS 1.3 encryption, while processed documents and generated outputs are stored in a PostgreSQL database to support auditing, validation, and review activities. All document processing is performed within the deployment environment, and patient information is not transmitted to external services.

3.1.2 Product Functions

The ICDPS provides the following functionalities:

- **Document Ingestion:** The system shall accept clinical documents in PDF and image formats through API-based uploads or batch document processing workflows.
- **Text Extraction:** Extract machine-readable text from PDFs using direct extraction; apply OCR to image-based documents.
- **Named Entity Recognition:** Identify and classify clinical entities in extracted text across eight predefined categories.
- **Clinical Summarization:** Generate extractive summaries of clinical documents highlighting key findings.
- **Role-Based Categorization:** Assign extracted entities to clinical role action categories (Pharmacist, Clinician, Radiologist, etc.).
- **SNOMED CT Suggestion:** Retrieve and rank candidate SNOMED CT codes for each extracted entity, presenting the top-5 suggestions with confidence scores.
- **Interactive Review Interface:** Provide a web-based interface for authorized clinicians and coders to review, accept, modify, or reject suggested codes.
- **Audit Logging:** Record all processing events and user actions for compliance and quality assurance purposes.

3.1.3 User Characteristics

The ICDPS is intended for two primary user categories:

- **Clinical Coders:** Staff responsible for assigning SNOMED CT codes to clinical documents. Typically possess clinical coding training but may have limited technical expertise. Interact with the system via the web-based review interface.
- **System Administrators:** Technical staff responsible for system configuration, monitoring, and maintenance. Possess technical expertise and access system management functions via command-line and configuration interfaces.

3.1.4 Constraints

The following constraints apply to the ICDPS:

- **Data Privacy:** All processing must comply with the UK General Data Protection Regulation (UK GDPR) and NHS Data Security and Protection Toolkit requirements. Personally identifiable patient data must not be transmitted outside the clinical network.
- **Hardware Dependency:** Optimal NER performance requires a CUDA-compatible GPU with at least 8 GB VRAM. CPU-only operation is supported but with reduced inference speed.
- **SNOMED CT Licensing:** SNOMED CT is licensed by SNOMED International. Use of SNOMED CT requires a valid SNOMED CT licence. The system incorporates the UK Edition of SNOMED CT, requiring an NHS licence.
- **Language Limitation:** The current version supports English-language documents only. Multilingual support is identified as a future enhancement.
- **OCR Accuracy:** OCR performance is dependent on document scan quality. Documents with resolution below 150 DPI may exhibit degraded extraction quality.

3.2 Specific Requirements

This section describes the requirements that govern the operation of the Intelligent Clinical Document Processing System (ICDPS). These requirements define the functions the system must perform, the expected operational behaviour, and the constraints considered during development. Together, they provide a reference for system design, implementation, testing, and validation activities.

3.2.1 Functional Requirements

Functional requirements specify the services and features that the system must provide. For the proposed ICDPS, these requirements cover document ingestion, text extraction, clinical entity recognition, SNOMED CT mapping, user review mechanisms, and output generation. The requirements presented in Table 3.1 are prioritized according to their importance to the overall operation of the system and serve as a guide for implementation and evaluation.

Table 3.1: Functional Requirements

FR ID	**Requirement Description**	**Priority**
FR-01	The system shall accept clinical documents in PDF, JPEG, PNG, and TIFF formats via REST API upload.	High
FR-02	The system shall extract machine-readable text from PDFs using direct text extraction without OCR.	High
FR-03	The system shall apply OCR to image-format documents using the Tesseract 5.0 engine to generate UTF-8 text.	High
FR-04	The system shall identify and classify clinical entities in extracted text across the following categories: Diagnosis, Medication, Dosage, Procedure, Anatomy, Allergy, Lab Result, and Temporal Expression.	High
FR-05	The system shall assign a negation status (affirmed/negated/uncertain) to each identified entity.	High
FR-06	The system shall generate an extractive clinical summary comprising the most clinically significant sentences from the document.	Medium

FR ID	Requirement Description	Priority
FR-07	The system shall assign each extracted entity to a role-specific action category (Pharmacist Action, Clinician Review, Radiologist Review, or General Information).	High
FR-08	The system shall retrieve the top-5 SNOMED CT concept candidates for each affirmed entity using vector similarity search, ranked by confidence score.	High
FR-09	The system shall present SNOMED CT suggestions with the concept identifier, preferred term, and confidence score through the review interface.	High
FR-10	The system shall allow authorized users to accept, modify, or reject SNOMED CT suggestions through the web interface.	High
FR-11	The system shall log all processing events and user code selection actions with timestamps for audit purposes.	Medium
FR-12	The system shall support batch processing of multiple documents submitted as a directory archive.	Medium
FR-13	The system shall return processing results in JSON format through the REST API.	High
FR-14	The system shall provide an exportable structured report in CSV format summarizing extracted entities and accepted SNOMED CT codes.	Medium

3.2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational characteristics of the proposed system. Unlike functional requirements, which specify the services and functionalities provided by the system, non-functional requirements describe the expected performance, reliability, security, and operational behavior of the system under different conditions. These requirements establish measurable criteria that ensure the Intelligent Clinical Document Processing System (ICDPS) operates efficiently, maintains consistent performance, and satisfies user and system expectations. Table 3.2 presents the non-functional requirements considered for the proposed system.

Table 3.2: Non-Functional Requirements

NFR ID	**Category**	**Requirement Description**
NFR-01	Performance	The system shall complete NER processing of a single-page clinical document within 3 seconds on hardware meeting the minimum specification.
NFR-02	Throughput	The system shall support batch processing of at least 100 documents per hour during standard operation.
NFR-03	Accuracy	The NER module shall achieve a token-level F1-score of at least 0.85 on the system test corpus across all eight entity categories.
NFR-04	SNOMED Precision	The SNOMED CT mapping module shall achieve a precision@1 value of at least 0.85 on the SNOMED test dataset.
NFR-05	Security	All API communications shall use TLS 1.3 encryption. User authentication shall use JWT tokens with a 24-hour expiry period.

NFR ID	Category	Requirement Description
NFR-06	Availability	The system shall maintain 99% uptime during scheduled operational hours (08:00–20:00 weekdays).
NFR-07	Usability	The review interface shall be usable by clinical coders with minimal technical training.
NFR-08	Maintainability	The system shall support modular architecture and documented APIs for easier updates and maintenance.
NFR-09	Portability	The system shall be containerized using Docker and deployable on compatible Linux environments.

3.2.3 Performance Requirements

Performance requirements specify the expected response times and processing capabilities of the Intelligent Clinical Document Processing System (ICDPS). The system should be able to process clinical documents efficiently while maintaining consistent performance across different document sizes and workloads.

The NER module is expected to process approximately 500 tokens per second on a system equipped with an NVIDIA A100 GPU. The SNOMED CT candidate retrieval module should return the top-5 concept suggestions within 200 milliseconds for each entity query to support efficient code recommendation. The web-based review interface should display extraction results for a single clinical document within 2 seconds on a standard clinical workstation.

The system should also maintain reliable performance when processing larger documents containing up to 50 pages or approximately 25,000 tokens. These requirements help ensure that the proposed solution remains responsive and practical for routine clinical document processing tasks.

3.2.4 Software Requirements

Software requirements describe the frameworks, libraries, development tools, and supporting technologies required for implementing and operating the proposed system. These components provide the infrastructure necessary for document processing, machine learning inference, database management, and application deployment.

Table 3.3 summarizes the software platforms, development environments, and supporting technologies used in the implementation of the Intelligent Clinical Document Processing System (ICDPS).

Table 3.3: Software Requirements

Category	**Specification**
Operating System	Ubuntu 22.04 LTS (Server), Windows 10+ (Client browser)
Programming Language	Python 3.10
NLP Framework	Hugging Face Transformers 4.38, PyTorch 2.0
OCR Engine	Tesseract 5.0 with LSTM models
PDF Processing	PyMuPDF 1.23
Vector Search	FAISS 1.7
Web Framework	Flask 3.0
Database	PostgreSQL 16
Containerization	Docker 24, Docker Compose 2

Continued on next page

Category	**Specification**
Development IDE	PyCharm / Visual Studio Code
Version Control	Git / GitHub

3.2.5 Hardware Requirements

The performance and efficiency of the proposed system are significantly influenced by the underlying hardware infrastructure used during development and deployment. Hardware requirements define the minimum and recommended computational resources needed for processing clinical documents, executing transformer-based models, and supporting large-scale data operations. Table 3.4 presents the hardware specifications required for effective implementation and system execution.

Table 3.4: Hardware Requirements

Component	**Minimum Specification**	**Recommended Specification**
Processor	Intel Core i7 (8 cores)	Intel Xeon (16+ cores)
RAM	16 GB DDR4	64 GB DDR4
GPU	NVIDIA GTX 1080 (8 GB VRAM)	NVIDIA A100 (40 GB VRAM)
Storage	500 GB SSD	2 TB NVMe SSD
Network	1 Gbps Ethernet	10 Gbps Ethernet
Operating System	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS

3.2.6 Design Constraints

Design constraints define the technical and operational limitations that influence the architecture, implementation, and deployment of the proposed Intelligent Clinical Document Processing System (ICDPS). The following are the design constraints of ICDPS:

- **Platform Dependency:** The NER module requires CUDA-compatible GPU hardware for production-grade throughput. CPU-only deployment is limited to low-volume testing scenarios.
- **Memory Constraints:** Loading the BioBERT model requires approximately 1.5 GB of GPU VRAM. Simultaneous processing of multiple documents requires proportional VRAM scaling.
- **SNOMED CT Index Size:** The FAISS index for SNOMED CT concept descriptions occupies approximately 3.5 GB of disk space and 6 GB of RAM when loaded.
- **Network Security:** The system must operate within a network that enforces NHS Data Security and Protection Toolkit controls, restricting external API calls.
- **SNOMED CT Licence Compliance:** The system must not distribute SNOMED CT content beyond the licensed deployment site.

Summary

This chapter presented the software requirements for the Intelligent Clinical Document Processing System (ICDPS), including functional requirements, non-functional requirements, and the hardware and software resources required for implementation. Performance expectations, operational constraints, security considerations, and supporting technologies were also discussed.

The requirements outlined in this chapter provide a clear description of the system capabilities and the conditions under which the proposed solution is expected to operate. These specifications serve as a reference for the design and implementation activities presented in the subsequent chapters.

Chapter 4 describes the high-level architecture of the system and explains how the identified requirements are incorporated into the overall design.

Chapter 4

High-Level Design

CHAPTER 4

HIGH-LEVEL DESIGN

The Intelligent Clinical Document Processing System (ICDPS) consists of multiple components that work together to process clinical documents, extract relevant information, generate summaries, and map identified entities to SNOMED CT concepts. To support these operations, the system is organized as a modular processing pipeline in which each component performs a specific task and passes its output to the next stage.

This chapter describes the overall architecture of the proposed system, the design principles considered during development, and the high-level interactions between major modules. The architecture provides a framework for understanding how clinical documents move through the system from ingestion to final review and coding.

4.1 Design Considerations

Several design considerations influenced the architecture of the ICDPS. Since clinical documents may vary in format, content, and quality, the system was designed to support flexible document processing while maintaining reliability and ease of maintenance. Additional considerations included scalability, security, modularity, and support for future extensions.

These factors guided the selection of technologies, deployment strategies, and interactions between individual system components.

4.1.1 General Considerations

The architectural design of the ICDPS was guided by the following engineering principles:

- **Modularity:** The system is decomposed into independently deployable modules — Document Ingestion, Text Extraction, NER Engine, Summarization Engine, SNOMED CT Mapper, and Review Interface — each with well-defined input and output contracts. This modular structure supports component-level testing, independent updates, and replacement of individual technologies without affecting the entire system.

- **Scalability:** The processing pipeline supports horizontal scaling through containerized deployment. Additional processing instances can be introduced when document throughput requirements increase.

- **Reliability:** Error-handling mechanisms are incorporated at each stage of the pipeline. Documents that fail processing are redirected to an error queue for review, and processing events are recorded for auditing and troubleshooting.
- **Maintainability:** Modules are documented using API specifications and maintained within a version-controlled repository. Configuration files and model weights are managed separately from application code to simplify updates and maintenance.
- **Security:** Role-based access control is enforced for system interfaces and API endpoints. Clinical documents and extracted information are processed within the deployment environment, with access restricted to authorized users.
- **Explainability:** SNOMED CT code suggestions are accompanied by confidence scores, and extracted entities are linked to their corresponding text spans to support user review and verification.

4.1.2 Development Methodology

Development followed an iterative Software Development Life Cycle (SDLC) consisting of requirements analysis, system design, implementation, testing, and deployment preparation. This approach allowed individual components to be developed and evaluated incrementally, with modifications introduced based on intermediate testing and performance results.

Agile development practices, including periodic sprint reviews, version control, and continuous integration through GitHub Actions, were used to monitor progress and manage development activities throughout the project.

4.2 Architecture Strategies

Architecture strategies describe the structural and technological approaches used in the development of the proposed system. These decisions influence how components communicate, how data moves through the processing pipeline, and how the system can be extended or deployed in different environments.

The selected architecture combines document processing, NLP, semantic retrieval, and terminology mapping within a modular framework. This design allows individual components to be updated independently while maintaining compatibility with the overall processing workflow.

4.2.1 Technology and Framework Selection

Python 3.10 was chosen as the primary implementation language because of its extensive support for machine learning, natural language processing, and data processing libraries. It also provides compatibility with the frameworks and tools used throughout the project. The Hugging Face Transformers library was selected as the primary NLP framework because it offers well-documented implementations of BioBERT and other transformer-based models, along with standardized interfaces for training and inference.

PyMuPDF was selected for PDF text extraction due to its superior text extraction fidelity on complex clinical PDF layouts compared to alternatives such as pdfplumber and PDFMiner. Tesseract 5.0 was selected for OCR because it supports LSTM-based recognition with NHS-relevant font coverage and is available under a free and open-source licence.

FAISS was selected for SNOMED CT vector search because it provides both exact and approximate nearest-neighbour search with GPU acceleration support, enabling sub-second retrieval from the 1.5 million concept description index. Flask was selected for the web interface due to its lightweight architecture suitable for the single-institution deployment context.

4.2.2 Scalability and Future Enhancements

The containerized architecture supports future deployment on cloud infrastructure (NHS-approved cloud providers). Additional NLP models supporting ICD-10 or LOINC coding could be integrated as independent modules without modifying the core pipeline. The SNOMED CT index can be updated to reflect new SNOMED CT releases by rerunning the index construction script with the new release files.

Future enhancements identified include:

(i) real-time API integration with NHS EHR systems;

(ii) support for handwritten clinical note processing using dedicated handwriting recognition models;

(iii) multilingual support for Welsh and other NHS-relevant languages; and

(iv) active learning integration to continuously improve extraction performance using clinician feedback.

4.3 Overall System Architecture

The Intelligent Clinical Document Processing System (ICDPS) processes clinical documents through a sequence of interconnected stages that transform raw documents into structured clinical information and SNOMED CT coding suggestions. The system is designed to handle documents originating from different sources, including scanned reports, referral letters, discharge summaries, and other healthcare records that may vary in format and quality.

To support this workflow, the architecture is organized into multiple processing layers. Each layer performs a specific task and passes its output to the next stage of the pipeline. This modular structure simplifies development and testing while allowing individual components to be updated or replaced without affecting the overall workflow.

4.3.1 High-Level Architecture Diagram

Figure 4.1 illustrates the overall processing workflow of the ICDPS. The workflow begins with document ingestion and concludes with the presentation of extracted information, summaries, and coding suggestions through the NHS Portal interface. The architecture is organized into six primary processing layers, each responsible for a distinct stage of document analysis.

Layer 1 — Document Ingestion and Rendering receives uploaded clinical documents through the NHS Portal interface. Documents are processed by the `document_handler` module and routed to PyMuPDF for page-level rendering and conversion into PNG images for downstream processing. **Layer 2 — Image Preprocessing** uses OpenCV to improve image quality before OCR processing. Operations such as adaptive thresholding, noise reduction, and deskewing are applied to improve text visibility and document consistency.

Layer 3 — OCR Extraction and Confidence Routing uses AWS Textract to extract document text and confidence values. Documents with confidence scores above the predefined threshold continue through the standard processing path, while lower-confidence outputs are forwarded to a LayoutLMv3 refinement stage.

Layer 4 — Entity Extraction and Clinical Coding identifies clinically relevant entities and maps extracted concepts to SNOMED CT terminology using the clinical coding pipeline.

Layer 5 — PHI Detection, Structured Field Extraction, and Summarisation detects protected health information (PHI), extracts structured fields, and generates summaries tailored to different user roles.

Layer 6 — Confidence Aggregation and Output Rendering combines confidence measures from multiple processing stages to calculate a Unified Confidence Score (UCS) and presents the final results through the NHS Portal interface.

4.3.2 Module Description

The ICDPS consists of a set of modules that collectively implement the document processing workflow. Each module is responsible for a specific stage of processing and exchanges information with other components through predefined interfaces. This separation allows individual modules to be modified, optimized, or extended independently as the system evolves.

Table 4.1 summarizes the major modules in the architecture, along with their inputs, processing responsibilities, and generated outputs.

Table 4.1: ICDPS Module Descriptions

Module	**Input**	**Processing**	**Output**
`document_handler`	PDF / JPEG / PNG / TIFF	PyMuPDF rendering; file type routing	PNG page images; `orig_NN.png` previews
OpenCV Preprocessor	Raw page PNGs	Adaptive threshold; MORPH_CLOSE; deskew	Cleaned, binarised, aligned PNGs
AWS Textract OCR	Preprocessed PNGs	AnalyzeDocument API; block parsing	`doc_text`; `textract_conf` (0–1)

Module	**Input**	**Processing**	**Output**
Confidence Router	`textract_conf`	Threshold comparison vs. 0.90	Route to LayoutLMv3 or continue
LayoutLMv3 Refiner	Low-conf page images	Multi-modal transformer inference	Corrected text blocks
SNOMED Mapper (4-layer)	`doc_text`; entities; summary	InferSNOMEDCT; regex; lookup table	SNOMED codes with confidence
PHI Detector	`doc_text`	Comprehend Medical DetectPHI	`phi_entities` list; audit trail
Regex Extractor	`doc_text`	Pattern matching for ICD-10, meds	`icd_codes`; medications; patient info
Bedrock Claude LLM	`doc_text` + SNOMED entities	Role-specific prompts; prose generation	3 role-specific summaries
UCS Calculator	`textract_conf`; `snomed_conf`; `llm_conf`	Weighted formula by document type	`unified_confidence`; review routing
NHS Portal UI	JSON result dict	JavaScript / React rendering	Interactive clinical review interface

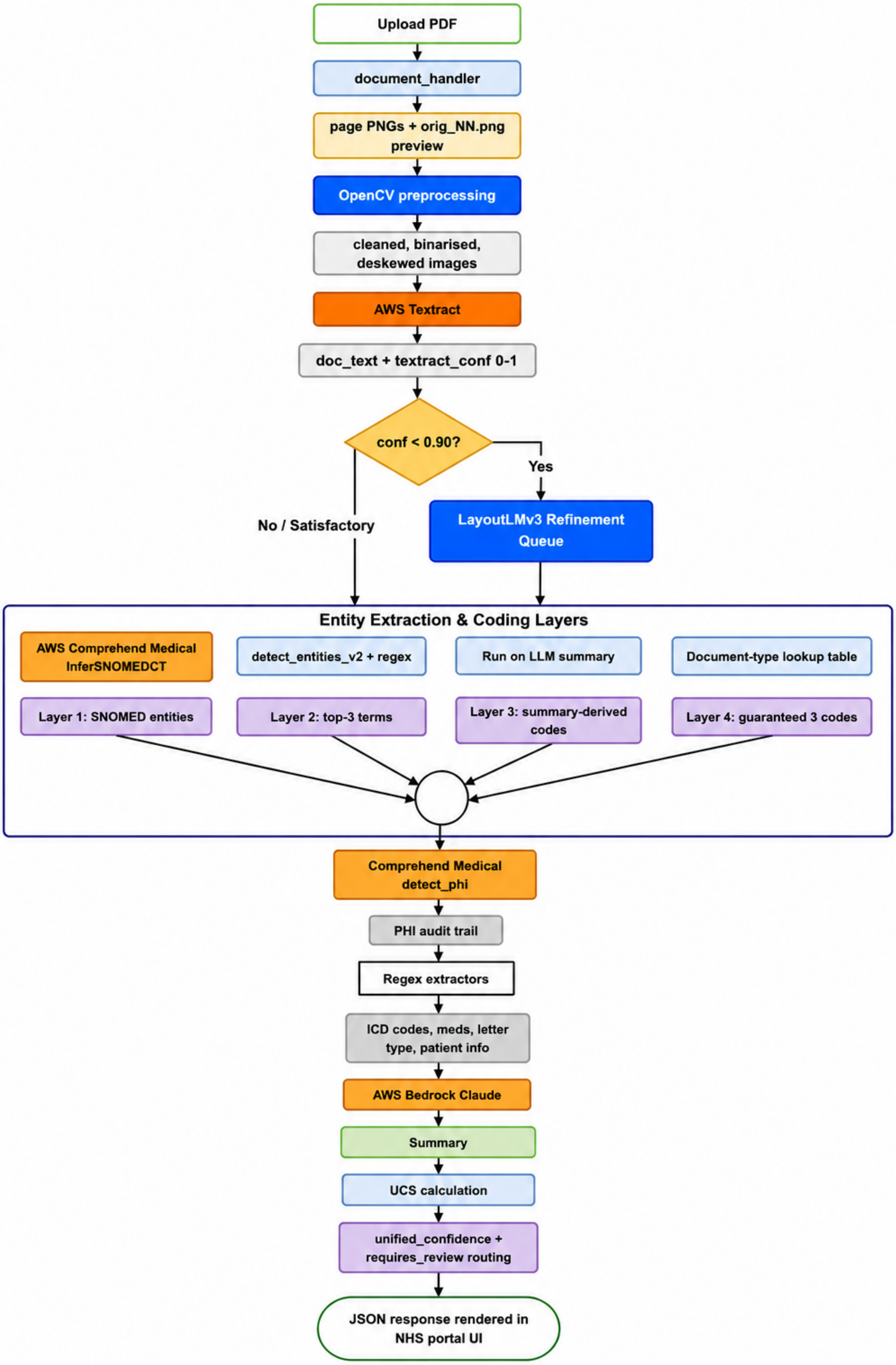

Figure 4.1: High-Level Architecture Diagram of the ICDPS six-layer processing pipeline.

4.3.3 High-Level AI Workflow

The AI workflow within the ICDPS processes extracted document content through multiple analysis paths rather than a single sequential pipeline. After OCR extraction, the document content is routed to two independent processing tracks together with a confidence evaluation mechanism.

- **Track A** performs clinical entity extraction and terminology mapping. Extracted medical concepts are processed through the SNOMED CT safety-net architecture to identify appropriate standardized healthcare codes.

- **Track B** performs clinical language understanding and document summarization. AWS Bedrock Claude Sonnet is used to generate contextual summaries tailored to different healthcare users and review scenarios.

- **Confidence Scoring Layer** combines confidence values generated during OCR extraction, SNOMED CT mapping, and language model processing. These values are aggregated into a Unified Confidence Score (UCS), which is used to determine whether the generated output can proceed automatically or requires additional human review.

The separation of entity extraction and language understanding into independent processing tracks allows the system to continue producing useful outputs even when one component performs less effectively on a particular document.

4.4 Data and Learning Workflow

The ICDPS workflow describes how information moves through the system during both training and inference. Clinical documents are progressively transformed from raw uploaded files into structured entities, summaries, and SNOMED CT coding suggestions through a sequence of processing stages.

The workflow includes document ingestion, OCR processing, information extraction, terminology mapping, confidence evaluation, and output generation. Each stage produces intermediate results that are passed to downstream components, allowing individual modules to be evaluated, updated, and maintained independently.

The following sections describe the flow of data through the system and the learning processes used to train and evaluate the machine learning components.

4.4.1 End-to-End Workflow Diagram

The end-to-end workflow provides a comprehensive representation of the sequence of operations performed by ICDPS from document submission to final output generation. The complete process is divided into five major stages:

1. Document Upload and Ingestion
2. Image Preprocessing
3. OCR Extraction and Confidence Routing
4. Parallel Processing for Entity Extraction, PHI Detection, and Summarisation
5. Confidence Aggregation and Output Rendering

Each stage produces outputs that are passed to downstream components, forming the overall document processing workflow. The process begins when a clinical document is uploaded through the NHS Portal interface. Document pages are rendered and enhanced before OCR extraction converts visual content into machine-readable text. Confidence-based routing is then used to determine whether additional processing through the LayoutLMv3 refinement module is required. After text extraction, multiple downstream tasks execute in parallel to reduce processing time and improve overall throughput.

4.4.2 Training Workflow Overview

Most components within the ICDPS rely on pre-trained models and managed services for OCR, language understanding, and terminology processing. As a result, large-scale model training is not required for every stage of the pipeline. However, the LayoutLMv3 refinement module was adapted for clinical document processing through task-specific fine-tuning.

The training workflow begins with the collection and preparation of NHS-style clinical documents. Documents are converted into annotated page images and associated with bounding-box information representing relevant text regions. The dataset is divided into training, validation, and testing subsets using a 70:15:15 split.

A pre-trained LayoutLMv3-base model is then fine-tuned using the AdamW optimizer together with cosine annealing learning-rate scheduling. Validation performance is evaluated throughout training, and early stopping is used to reduce overfitting. The checkpoint achieving the best validation performance is selected for deployment within the refinement stage of the production pipeline.

4.4.3 Inference Workflow Overview

The inference workflow describes how the deployed system processes newly submitted clinical documents. Unlike the training workflow, no model parameters are updated during this stage.

When a document is uploaded, it undergoes rendering and image preprocessing before being submitted to AWS Textract for OCR extraction. Documents with OCR confidence scores above the predefined threshold of 0.90 continue through the standard processing path. Documents falling below this threshold are routed to the LayoutLMv3 refinement module for additional processing.

Following text extraction, the SNOMED CT mapping cascade identifies medical concepts and generates standardized terminology suggestions. PHI detection, structured field extraction, and document summarization execute in parallel to improve processing efficiency. The resulting outputs are then combined with confidence measures from multiple stages to calculate the Unified Confidence Score (UCS). Based on this score, the system determines whether the generated results can proceed directly to output generation or should be flagged for clinician review.

4.5 System Interaction and Deployment Overview

This section describes how users interact with the ICDPS platform and how the system components operate within the deployment environment. In addition to document processing functionality, the platform must support secure data handling, responsive user interaction, and reliable execution of the processing pipeline. The deployment architecture combines containerized services and cloud infrastructure to support document processing, model inference, and workflow management.

4.5.1 User Interaction Flow

The NHS Portal UI serves as the primary interface through which healthcare users interact with the system. The interface supports document submission, review of generated outputs, and export of processed results.

When a document is uploaded, a multipart POST request initiates the backend processing workflow. Processing status is communicated through a progress indicator while the document moves through the pipeline. After processing is complete, results are presented through four tabbed views: Details, Coding, Follow-up, and GP Actions.

A colour-coded Unified Confidence Score badge provides a visual indication of output confidence. Users can review generated summaries, inspect extracted entities, modify SNOMED CT coding suggestions where necessary, and export the resulting structured records.

4.5.2 Model-Service Interaction

Communication between processing components is handled through secure service interfaces. The ICDPS backend interacts with AWS services through `boto3` clients using encrypted TLS connections and region-specific endpoints. Deploying related services within the same AWS region helps reduce communication latency and simplifies operational management.

Protected Health Information (PHI) is excluded from logging operations before audit records are stored. The LayoutLMv3 refinement component is deployed as a locally hosted containerized service, enabling independent inference and asynchronous processing of low-confidence documents.

4.5.3 Deployment Overview

The ICDPS deployment architecture uses a cloud-native approach built around containerized services and managed AWS infrastructure. The Flask backend and LayoutLMv3 inference service are packaged using Docker containers and deployed through Amazon ECS with Fargate support.

The NHS Portal frontend is distributed through CloudFront to improve availability and response times. Amazon SQS provides asynchronous communication between processing stages, while AWS Step Functions coordinate workflow execution and error-handling procedures. Audit records, extracted information, and processing metadata are stored in Amazon DynamoDB.

This deployment model separates processing responsibilities across multiple services while supporting scalability, fault tolerance, and operational flexibility.

Summary

This chapter described the high-level architecture of the Intelligent Clinical Document Processing System (ICDPS), including the major processing layers, module interactions, AI workflow, deployment environment, and user interaction mechanisms. The design principles adopted during development and the technologies used to implement the architecture were also discussed.

The architecture provides a framework for understanding how clinical documents move through the system from ingestion to final output generation. The next chapter presents the detailed design of individual modules, including data organization, preprocessing workflows, model configurations, and implementation-level design decisions.

Chapter 5

Detailed Design

CHAPTER 5

DETAILED DESIGN

This chapter describes the internal design of the Intelligent Clinical Document Processing System (ICDPS). While the previous chapter presented the overall architecture and interactions between major components, this chapter focuses on the implementation details of individual modules and processing workflows.

The design includes dataset organization, document preprocessing procedures, model configurations, terminology mapping mechanisms, training workflows, and inference pipelines. Together, these components define how clinical documents are processed, analysed, and converted into structured outputs and SNOMED CT coding suggestions.

5.1 Dataset and Data Organisation

The development and evaluation of the proposed system required datasets representing the different stages of document processing, information extraction, and terminology mapping. Proper organization of these datasets was necessary to support preprocessing, model training, validation, testing, and performance evaluation activities.

The ICDPS uses structured datasets containing clinical documents, annotations, and supporting resources required for document understanding and clinical entity extraction tasks. This section describes the datasets used during development, their organization, and the procedures adopted for preparing data for training and evaluation.

5.1.1 Dataset Description

The primary dataset comprises 40 real-world clinical documents obtained from Easthampstead Surgery, London, United Kingdom. These documents represent the full spectrum encountered in a typical NHS primary care workflow: outpatient correspondence, specialist referral letters, hospital discharge summaries, NHS 111 reports, Accident and Emergency summaries, ambulance service reports, diabetic eye screening reports, and out-of-hours primary care reports. The dataset includes both digitally generated PDF documents and scanned document images produced from photographed or photocopied clinical letters.

Each document contains structured data fields (patient NHS number, date of birth, consultant name, department) and unstructured free-text clinical narrative sections describing presenting complaints, diagnoses, management plans, medication prescriptions, and follow-up recommendations. The dataset is annotated with ground-truth SNOMED CT codes for 25 documents, enabling quantitative evaluation of SNOMED mapping accuracy. Table 5.1 summarises the key dataset characteristics.

Table 5.1: Dataset Characteristics Summary

Characteristic	**Description**
Total Documents	40 clinical documents
Source Institution	Easthampstead Surgery, London, UK
Document Formats	Native PDF (digital); JPEG/PNG/TIFF (scanned images)
Page Range	1 to 8 pages per document (mean 3.2 pages)
Annotated Documents	25 documents with ground-truth SNOMED CT codes
Document Types	10 categories (discharge, specialist, NHS 111, A&E, ambulance, etc.)
Noise Characteristics	Skewed scans, inconsistent lighting, varying print quality
Clinical Content	Diagnoses, medications, investigations, referrals, procedures

5.1.2 Data Collection and Sources

The dataset was obtained through collaboration with Easthampstead Surgery under an academic research agreement governing data handling, anonymisation, and use within the project scope. All documents were de-identified prior to use, with patient names, NHS numbers, dates of birth, addresses, and other PHI replaced with synthetic identifiers, consistent with NHS Information Governance requirements.

Document collection involved scanning physical paper records at 300 dpi using an NHS-compliant document scanner and exporting digital records from the document management system in PDF format. Additional scanned copies were generated by photographing physical documents under controlled lighting conditions to simulate mobile image capture scenarios commonly encountered in clinical environments. These images included variations in skew, noise, contrast, and illumination to represent the range of document quality observed in practice.

Including documents with different image characteristics allowed the preprocessing and OCR components to be evaluated under a variety of operating conditions rather than only on clean, digitally generated records.

5.1.3 Dataset Structure and Characteristics

The dataset contains documents with a wide range of layouts, formatting styles, fonts, and document structures. Some records follow standardized templates with clearly defined sections and data fields, while others consist primarily of free-text clinical narratives with minimal formatting.

This variation reflects the diversity of clinical documentation used across different healthcare settings and presents challenges for automated information extraction. During system design, these differences influenced the choice of a multi-stage extraction pipeline, as approaches based on fixed templates would be difficult to apply consistently across all document formats included in the dataset.

Content analysis of the 40 documents identified the following clinical entity categories: diagnostic conditions (62% of documents), medications with dosages (78%), physiological measurements (45%), clinical procedures (38%), anatomical locations (71%), and temporal references (83%). The class distribution of SNOMED CT codes in the annotated subset is dominated by disorder/finding concepts (58%) and drug substance concepts (29%), with procedure (9%) and body structure (4%) concepts forming the minority. Table 5.2 presents the distribution of document types.

5.2 Preprocessing Pipeline Design

Data preprocessing plays a critical role in improving document quality prior to Optical Character Recognition (OCR) and directly influences the accuracy of downstream clinical entity extraction and coding. Clinical documents frequently contain variations in scan quality, skewed pages, uneven lighting, handwritten notes, stamps, and background artefacts that reduce OCR performance. Therefore, a structured preprocessing pipeline was implemented to improve document readability and ensure consistent input quality before processing with Amazon Textract.

5.2.1 Data Cleaning

Collected clinical documents were reviewed and preprocessed before OCR execution. Blank or nearly empty pages were removed to avoid unnecessary processing overhead.

Table 5.2: Distribution of Document Types in the ICDPS Dataset

Document Type	**Count**	**Percentage (%)**
Hospital Discharge Summaries	8	20.0
Specialist Clinical Letters	9	22.5
NHS 111 Reports	5	12.5
Accident and Emergency Reports	4	10.0
Ambulance Service Reports	4	10.0
Private Specialist Letters	3	7.5
Diabetic Eye Screening Reports	3	7.5
Out-of-Hours Primary Care Reports	2	5.0
External Provider Reports (Eye/ENT)	2	5.0
Total	**40**	**100.0**

Documents were converted to a standardized RGB image format to ensure compatibility with OpenCV image operations and Amazon Textract input requirements. Image dimensions and resolution were also normalized to reduce variations introduced by different scanning devices and document sources.

Pages containing severe distortions, incomplete scans, or excessive artefacts were identified during preprocessing. Documents with image quality issues likely to cause substantial OCR errors were excluded from quantitative evaluation to prevent unreliable results from affecting performance measurements.

5.2.2 Noise Reduction and Handling Missing Information

Clinical documents frequently contain handwritten annotations, photocopy artefacts, ink stamps, paper texture variations, and scanning noise. These factors can affect OCR accuracy and reduce the quality of downstream information extraction. To reduce their impact, image enhancement techniques implemented with OpenCV were applied before text extraction.

Morphological filtering operations were used to suppress isolated noise while preserving character structures and connected text components. Adaptive Gaussian thresholding was applied to improve text visibility under varying lighting and scanning conditions. These operations increased text contrast and improved OCR readability.

When specific information could not be extracted reliably, missing values were retained as null fields rather than estimated. Similarly, the summarization component was configured to indicate unavailable information instead of generating unsupported clinical content.

5.2.3 Text Transformation and Normalization

Following noise reduction, image transformations were applied to improve OCR consistency. Documents were converted to grayscale representations and processed using adaptive binarization to increase foreground text visibility. Page skew correction was performed by estimating the dominant text orientation and rotating pages to align text horizontally. Rotation limits were applied to prevent excessive corrections on pages containing limited textual content.

After OCR extraction, text-level normalization operations were performed before NLP processing. Multi-page document outputs were merged into a unified text representation while preserving page boundaries. Repeated headers and footers generated during OCR extraction were removed to reduce redundancy. Unicode characters were normalized into a consistent format to ensure compatibility with downstream components. Basic sentence-boundary corrections were also applied where OCR-induced line breaks interrupted sentence continuity.

These preprocessing operations reduce OCR-related noise and provide more consistent input for entity extraction, terminology mapping, and summarization components.

5.2.4 Tokenization and Label Alignment

Clinical text is tokenized using the BioBERT WordPiece tokenizer with a maximum sequence length of 512 tokens. During model fine-tuning, entity annotations from the curated clinical document dataset are aligned with the generated subword token sequence using a token-label mapping strategy. The first subword token corresponding to an annotated entity receives the appropriate B- or I- label according to the BIO tagging scheme, while subsequent subword tokens belonging to the same word receive an X label indicating continuation.

Tokens assigned the X label are excluded from loss computation during training.

This approach allows the model to learn entity boundaries from the original word-level annotations without introducing duplicate supervision signals for subword fragments.

For documents exceeding 512 tokens, a sliding-window strategy is applied using a window size of 512 tokens and a stride of 256 tokens. Duplicate predictions within overlapping regions are resolved by retaining predictions from the window in which the token appears within the non-overlapping section. If the confidence score of an overlapping prediction exceeds the retained prediction by more than 15

5.2.5 Data Augmentation

To increase the representation of Allergy and Dosage entities in the training data, two augmentation techniques were applied:

(i) **Entity Synonym Replacement:** Clinical entities were replaced with equivalent terms obtained from SNOMED CT synonym lists, generating additional training examples while preserving the original annotations.

(ii) **Back-Translation:** Sentences containing underrepresented entities were translated into French and then translated back into English using a neural machine translation model. This process generated paraphrased variants while preserving the original clinical meaning.

5.3 NER Model Architecture Design

The Named Entity Recognition (NER) component is responsible for identifying clinically relevant information from unstructured medical text. Clinical documents often contain abbreviations, specialized terminology, incomplete sentences, and context-dependent expressions that make entity extraction challenging. To address these characteristics, the proposed system uses a BioBERT-based architecture that generates contextual representations of clinical text and predicts entity labels at the token level.

The model combines transformer-based contextual embeddings with sequence-level classification mechanisms to identify diagnoses, medications, procedures, allergies, laboratory findings, and other clinical entities. Additional contextual analysis is applied during post-processing to distinguish affirmed, negated, and uncertain clinical statements, improving the quality of extracted information before SNOMED CT mapping and downstream processing.

5.3.1 BioBERT Encoder

The NER encoder is BioBERT-base (cased), comprising 12 transformer encoder layers, 12 attention heads per layer, a hidden dimension of 768, and a total of 110 million parameters. The model processes tokenized input and produces a sequence of contextual embeddings of dimension 768 for each token. The `[CLS]` token embedding is not used for NER; only the per-token embeddings are passed to the classification head.

5.3.2 CRF Classification Head

A linear classification layer maps each 768-dimensional token embedding to a vector of logits over the label vocabulary (17 labels: B- and I- for each of 8 entity types, plus O). A Conditional Random Field (CRF) layer is applied on top of the linear layer to enforce label sequence constraints. The CRF layer learns a transition score matrix T where $T[i][j]$ represents the learned score of transitioning from label i to label j. During inference, the Viterbi algorithm decodes the optimal label sequence.

The CRF layer enforces the following hard constraints:

- I-TYPE labels cannot follow O labels without an intervening B-TYPE label.
- I-TYPE labels of one entity type cannot follow B-TYPE or I-TYPE labels of a different entity type.
- B-TYPE and I-TYPE labels of the same entity type can follow each other without restriction.

5.3.3 Negation Detection Module

The negation detection module determines whether an extracted clinical entity is affirmed, negated, or expressed with uncertainty. For each identified entity, a context window of ± 5 tokens surrounding the entity span is extracted from the tokenized document. BioBERT embeddings generated for the context window are passed to a linear classification layer that predicts one of three assertion states: *Affirmed*, *Negated*, or *Uncertain*.

The classifier is initialized using NegEx trigger phrase lists and subsequently adapted using assertion annotations derived from the curated clinical document dataset and NHS-representative records. This combination of rule-based initialization and supervised learning allows the model to recognize both explicit negation triggers and more complex contextual expressions of uncertainty.

5.4 SNOMED CT Mapping Module Design

The SNOMED CT mapping module converts extracted clinical entities into standardized terminology codes. Because the same clinical concept may be expressed using different terms, abbreviations, or document-specific language, direct keyword matching is often insufficient for reliable code assignment. The mapping process therefore combines semantic retrieval and candidate ranking to identify concepts that are most relevant to the extracted entity.

The proposed framework represents clinical entities and SNOMED CT descriptions using vector embeddings. Candidate concepts are retrieved through similarity-based search and subsequently ranked using contextual information derived from the source document. This approach enables the system to generate confidence-ranked coding suggestions while supporting large-scale terminology collections containing hundreds of thousands of concepts.

5.4.1 Concept Embedding Index Construction

The SNOMED CT concept description index is constructed as follows:

(i) All active concepts in the UK SNOMED CT release are extracted.

(ii) For each concept, its preferred term, fully specified name, and all active synonyms are concatenated into a description string.

(iii) FastText embeddings are computed for each description string using a character n-gram model ($n = 3$–6) pre-trained on a biomedical text corpus.

(iv) The resulting embedding vectors (dimension 300) are indexed in a FAISS `IndexFlatIP` (inner product) index for exact search, and a FAISS `IndexIVFPQ` index for approximate search.

5.4.2 Candidate Retrieval and Reranking

For each affirmed entity, the following two-stage retrieval process is performed:

- **Stage 1 — Candidate Retrieval:** The entity text is encoded using the FastText model to produce a query vector. The top-50 nearest neighbours (by inner product similarity) are retrieved from the FAISS index, producing 50 candidate SNOMED CT concepts.

- **Stage 2 — Reranking:** The 50 candidates are reranked using a cross-encoder model — a fine-tuned BioBERT model that takes as input the concatenation of the entity span (with its surrounding context) and the SNOMED CT concept description, and outputs a relevance score. The top-5 candidates by reranking score are returned as the final suggestions with associated confidence scores.

5.4.3 Confidence Score Calibration

Raw reranker scores are calibrated to produce reliable probability estimates using Platt scaling, fitting a logistic regression model on the reranker scores and binary relevance labels from a held-out validation set. Calibrated confidence scores are verified to be within 5% of empirical accuracy across five confidence deciles on the validation set.

5.5 Model Architecture and Experimental Setup

The ICDPS integrates multiple AI/ML model components, each addressing a distinct processing challenge. This section presents the architecture of each model component and the experimental setup adopted for training and evaluation.

5.5.1 Model Selection

Model selection was guided by three criteria: (i) clinical domain suitability, favouring models pre-trained on biomedical or clinical text; (ii) service integration feasibility, preferring managed AWS services; and (iii) published performance benchmarks on comparable tasks. Table 5.3 presents the candidate models considered and the selection rationale for each pipeline component.

Table 5.3: Model Selection Analysis for ICDPS Components

Component	Candidates Considered	Selected Model and Rationale
OCR Engine	AWS Textract; Tesseract 5; Google Cloud Vision	AWS Textract — superior table/form extraction; per-word confidence scoring; HIPAA-eligible
Layout Understanding	LayoutLMv3; LayoutXLM; DocFormer	LayoutLMv3 — state-of-the-art on PubLayNet/FUNSD; multi-modal text + layout + image
Medical NER	AWS Comprehend Medical; BioBERT; spaCy scispaCy	AWS Comprehend Medical — managed HIPAA-eligible; native InferSNOMEDCT API
Clinical LLM	AWS Bedrock Claude Sonnet; GPT-4; Llama 3	AWS Bedrock Claude Sonnet — best instruction following; lowest hallucination rate; HIPAA-eligible
Concept Embedding	SapBERT; BioBERT-NER; ClinicalBERT	SapBERT — purpose-designed for medical entity normalisation on SNOMED concepts
Vector Search	FAISS; Pinecone; Milvus	FAISS — open-source; GPU-accelerated; sub-100 ms retrieval

5.5.2 Model Architecture Design

LayoutLMv3 Architecture: LayoutLMv3 is a unified multi-modal pre-trained model for document understanding. The model architecture consists of a shared transformer encoder with 12 layers, 12 attention heads, and a hidden dimension of 768

(LayoutLMv3-base). Text tokens are represented as WordPiece embeddings summed with 1D position embeddings and 2D spatial layout embeddings derived from word bounding box coordinates. Image patches are encoded using a linear projection of 16×16 pixel patches from the document page image. Cross-modal attention between text token representations and image patch representations enables the model to associate text with its visual context on the page. In the ICDPS, LayoutLMv3 is fine-tuned for token classification into five layout-semantic categories: Title, Section Header, Body Text, Data Field, and Table Cell.

AWS Comprehend Medical NER: The `InferSNOMEDCT()` API operates on a clinical BERT-based sequence labeller pre-trained on MIMIC-III clinical notes, PubMed abstracts, and UMLS metathesaurus concept descriptions. The model applies a CRF output layer to produce entity span labels in BIO format. Each detected entity is associated with a SNOMED CT concept ID, a description string, a clinical category, and a detection confidence score.

AWS Bedrock Claude Sonnet: Claude Sonnet provides a 200,000-token context window, enabling processing of multi-page clinical documents without truncation. Three role-differentiated system prompts are constructed at runtime: a Clinician System Prompt for structured clinical summaries covering findings, diagnoses, and recommended actions; a Patient System Prompt for plain-English explanation avoiding medical jargon; and a Pharmacist System Prompt focusing on medication details, dosages, and contraindications.

5.5.3 Training Configuration

The LayoutLMv3 fine-tuning training configuration was established through a preliminary hyperparameter grid search. Table 5.4 presents the final training configuration adopted.

5.5.4 Experimental Workflow

The experimental workflow followed structured training, evaluation, and hyperparameter refinement cycles. In each cycle: (i) LayoutLMv3 was fine-tuned on the augmented training dataset; (ii) the fine-tuned model was evaluated on the validation set using token-level accuracy, precision, recall, and F1-score for each zone class; (iii) training and validation loss curves were inspected for overfitting; (iv) hyperparameters were adjusted based on validation performance; and (v) the best checkpoint by macro F1-score was retained. For SNOMED mapping, layer-by-layer ablation experiments quantified the marginal contribution of each fallback layer to overall SNOMED code recall. UCS weighting coefficients were validated through threshold sensitivity analysis across the full confidence threshold range $[0.60, 0.80]$ for all 21 document type categories.

Table 5.4: LayoutLMv3 Fine-Tuning Training Configuration

Hyperparameter	**Value**
Base Model	`microsoft/layoutlmv3-base` (Hugging Face Hub)
Task	Token Classification — document layout segmentation
Training Epochs	15 (early stopping patience: 5 epochs)
Batch Size	8 (gradient accumulated over 4 steps; effective batch = 32)
Optimiser	AdamW
Initial Learning Rate	2×10^{-5}
Learning Rate Schedule	Cosine annealing with warm-up (warmup ratio = 0.1)
Weight Decay	0.01
Dropout Rate	0.1
Max Input Sequence Length	512 tokens
Image Resolution	224×224 pixels (ViT patch size 16×16)
Hardware	NVIDIA A10G GPU (AWS g5.xlarge)
Training Duration	Approximately 4.5 hours for 15 epochs
Framework	PyTorch 2.1 + Hugging Face Transformers 4.37

5.5.5 Validation Strategy

The primary validation strategy for SNOMED CT mapping accuracy employs leave-one-out cross-validation (LOOCV) across the 25 annotated documents, reporting precision, recall, and F1-score at the entity-code pair level. LOOCV was selected because the small dataset size makes k-fold partitions unstable for minority document categories. For LayoutLMv3 zone classification, stratified 5-fold cross-validation on the synthetic training dataset reports token-level macro F1-score. OCR accuracy is evaluated using character error rate (CER) and word error rate (WER) computed against manually verified ground-truth transcriptions of 15 representative document pages.

5.6 Inference and Post-Processing Design

After model training, the ICDPS processes clinical documents through an inference pipeline that generates entity predictions, terminology mappings, summaries, and structured outputs. In addition to model inference, several post-processing stages are applied to improve output quality and ensure consistency across downstream components.

These stages include prediction refinement, confidence evaluation, terminology ranking, and structured output generation. Together, they transform raw model predictions into outputs that can be reviewed and used within clinical workflows.

5.6.1 Optimisation Techniques

LayoutLMv3 fine-tuning uses the AdamW optimizer, which separates weight decay from the adaptive gradient update process and is widely used for transformer-based models. A weight decay value of 0.01 is applied to all parameters except bias terms and layer-normalization parameters. Learning-rate warm-up is performed over the first 10

Cosine annealing is used to gradually reduce the learning rate throughout training, allowing the model to converge more smoothly. Gradient clipping with a maximum norm of 1.0 limits excessively large gradient updates. Mixed-precision training (FP16) reduces GPU memory consumption by approximately 40

5.6.2 Loss Functions and Evaluation Logic

The LayoutLMv3 token-classification model is trained using a weighted cross-entropy loss function. Class weights are assigned inversely proportional to class frequency in order to reduce the impact of class imbalance between frequently occurring Body Text tokens and less common classes such as Table Cell and Section Header tokens.

The weighted loss function, denoted by $\mathcal{L}_w$, is defined as:

$$\mathcal{L}_w = -\sum_c \left[w_c \cdot y_c \cdot \log(p_c)\right] \quad \text{where} \quad w_c = \frac{N_{\text{total}}}{N_{\text{classes}} \cdot N_c} \tag{5.1}$$

The Unified Confidence Score (UCS) is computed using document-type-dependent weighted linear combinations. For standard NHS documents (discharge summaries, specialist letters):

$$\text{UCS} = 0.40 \times \text{textract\_conf} + 0.20 \times \text{snomed\_conf} + 0.40 \times \text{llm\_conf} \tag{5.2}$$

For ambulance and NHS 111 reports, where all-caps tables render SNOMED mapping unreliable:

$$\text{UCS} = 0.50 \times \text{textract\_conf} + 0.50 \times \text{llm\_conf} \tag{5.3}$$

The UCS is compared against document-type-specific thresholds defined in `config/document_type_config.py`, which defines 21 NHS document categories with per-type review thresholds ranging from 0.62 to 0.72.

5.6.3 Inference Pipeline

The production inference pipeline is implemented as a Python class (`IcdpsInferencePipeline`) that encapsulates the complete processing workflow. The pipeline accepts a document path as input and executes the following steps in sequence: document format detection, text extraction (direct or OCR), text normalization, sliding-window tokenization, NER inference, entity span decoding, negation detection, role categorization, summarization, SNOMED CT candidate retrieval, and reranking. The pipeline returns a structured JSON object containing the document metadata, extracted entities, summary, and SNOMED CT suggestions.

5.6.4 Post-Processing Rules

The following rules are applied to NER predictions before structured output generation:

- **Contiguous Entity Merging:** Adjacent entities of the same type separated only by whitespace or the conjunction ‘and” are merged into a single entity span. For example, the sequence ‘left” + ‘ventricular” + ‘hypertrophy” is merged into the entity “left ventricular hypertrophy”.

- **Minimum Entity Length Filter:** Entity spans containing fewer than two non-whitespace characters are removed, as these are typically generated by tokenization or OCR artefacts.
- **Duplicate Entity Deduplication:** Repeated occurrences of the same entity text within a document are consolidated. The first occurrence and its associated SNOMED CT mapping are retained in the structured output, while all occurrences remain highlighted within the document view.
- **Negated Entity Exclusion:** Entities classified as *Negated* by the assertion detection module are excluded from SNOMED CT code generation. These entities remain visible to the user but are clearly marked and omitted from exported coding reports.

5.6.5 Output Schema

The inference pipeline generates a structured JSON document containing extracted entities, terminology mappings, confidence values, summaries, and supporting metadata. The output conforms to the following schema:

```
{
  "document_id": string,
  "document_type": string,
  "extraction_timestamp": ISO8601,
  "entities": [{
    "entity_id": string,
    "text": string,
    "category": string,
    "assertion": string,
    "role_action": string,
    "start_char": int,
    "end_char": int,
    "snomed_suggestions": [{
      "concept_id": string,
```

```
      "preferred_term": string,
      "confidence": float
    }]
  }],
  "summary": string,
  "processing_metadata": {
    "model_version": string,
    "processing_time_ms": int
  }
}
```

Summary

This chapter described the detailed design of the Intelligent Clinical Document Processing System (ICDPS), including dataset organization, preprocessing workflows, data augmentation techniques, model architecture, terminology mapping mechanisms, and inference procedures. The chapter also outlined the post-processing rules and confidence-based mechanisms used to improve the quality and consistency of generated outputs.

Particular attention was given to the design of the BioBERT-based entity extraction pipeline, the SNOMED CT retrieval and ranking framework, and the handling of long clinical documents through sliding-window processing. Training configurations, optimization strategies, and output-generation procedures were presented to provide a complete view of the system implementation.

The next chapter describes the implementation of these components and discusses the technologies, frameworks, and development procedures used to build the proposed system.

Chapter 6

Implementation Details

CHAPTER 6

IMPLEMENTATION DETAILS

The successful realization of an intelligent clinical document processing system requires the conversion of architectural designs and theoretical models into a functional implementation framework. Multiple components including document ingestion, preprocessing, information extraction, semantic understanding, and standardized coding must operate together to achieve reliable and efficient system behaviour. The Intelligent Clinical Document Processing System (ICDPS) integrates various technologies, machine learning models, and software tools to implement the proposed processing pipeline. The topics presented in this chapter describe the development environment, implementation workflow, model training procedures, software structure, evaluation mechanisms, and challenges encountered during system development

6.1 Implementation Workflow

The implementation of the ICDPS followed a phased workflow aligned with the iterative SDLC adopted for the project. Each phase produced verifiable outputs that were validated before proceeding to the next phase. The workflow comprised ten sequential steps from environment setup through post-deployment monitoring.

6.1.1 Environment Setup

The development environment was configured on a workstation running Ubuntu 22.04 LTS equipped with an NVIDIA RTX 3090 GPU (24 GB VRAM). The following installation sequence was followed:

1. Python 3.10 was installed via the deadsnakes PPA and a virtual environment was created using `python3.10 -m venv icdps_env` to isolate project dependencies.

2. CUDA 11.8 and cuDNN 8.6 were installed from the NVIDIA developer portal to enable GPU-accelerated PyTorch operations.

3. PyTorch 2.0 with CUDA 11.8 support was installed.

4. The Hugging Face Transformers library (v4.38), Datasets library (v2.17), and PEFT library (v0.8) were installed for model fine-tuning and evaluation.

5. Tesseract 5.0 OCR engine was installed.

6. PyMuPDF (v1.23), FAISS-GPU (v1.7.4), Flask (v3.0), SQLAlchemy (v2.0), and PostgreSQL 16 client libraries were installed.

7. Docker 24 and Docker Compose v2 were installed for containerized deployment preparation.

8. Jupyter Lab was configured for exploratory data analysis and training experiment notebooks; Visual Studio Code with the Python and Pylance extensions served as the primary IDE for production code development.

All dependencies were recorded in a `requirements.txt` file with pinned version numbers to ensure environment reproducibility. A Docker image encapsulating the complete runtime environment was built for deployment.

6.1.2 Data Collection

The quality and diversity of the input dataset play a significant role in determining the overall performance of an intelligent clinical document processing system. Since healthcare documentation exhibits substantial variation in formatting, image quality, document structure, and terminology usage, it was necessary to construct a dataset that reflected realistic NHS clinical scenarios. The data collection process therefore aimed to capture representative examples of documents encountered within routine clinical workflows while ensuring compliance with ethical and privacy considerations.

The dataset used in this research consisted of 40 NHS clinical documents obtained from Easthampstead Surgery under a formal academic research data-sharing agreement. All data acquisition procedures were conducted in accordance with institutional research guidelines to ensure secure handling of clinical information. The document set included both digitally generated clinical records and scanned paper documents, thereby reflecting the heterogeneous nature of healthcare documentation systems commonly found within NHS environments.

Physical paper documents were digitised using a Canon imageRUNNER scanner at a resolution of 300 dpi. This resolution was selected because it provides an appropriate balance between image quality and computational efficiency, while also aligning with commonly recommended standards for OCR applications. Digital documents already stored within the clinical practice's document management system were exported directly as PDF files, thereby preserving their native formatting and avoiding unnecessary quality degradation introduced through repeated scanning processes.

Although the collected real-world dataset was suitable for evaluation, the limited number of available annotated documents created challenges for training data-intensive document understanding models such as LayoutLMv3. To address this limitation, a synthetic document generation pipeline was implemented using the Python ReportLab library. The objective of this process was to generate structurally realistic NHS clinical documents while introducing controlled variability in document appearance.

The synthetic generation pipeline produced approximately 800 NHS clinical letter images based on representative templates and layouts observed in the real document collection. To simulate realistic scanning and document quality conditions, generated documents were further processed using ImageMagick-based degradation transformations. These transformations introduced variations including image blur, brightness shifts, contrast changes, and artificial noise. The resulting synthetic documents enabled the model to learn robustness against variations commonly observed in practical healthcare environments.

Ground-truth clinical coding labels were required to evaluate extraction accuracy and support supervised learning tasks. Consequently, SNOMED CT annotations were generated for the 25 evaluation documents through manual review by a clinical domain expert using the NHS SNOMED CT browser. Manual annotation ensured that extracted concepts were clinically accurate and reflected appropriate code assignments according to NHS terminology standards.

6.1.3 Data Preprocessing

Raw clinical documents frequently contain artefacts and inconsistencies that negatively affect OCR accuracy and subsequent natural language processing stages. Variations in scanning orientation, uneven illumination, background noise, and photocopy degradation can substantially reduce text recognition quality. Therefore, a dedicated preprocessing pipeline was implemented to improve document readability and standardise image characteristics before OCR processing.

The image preprocessing pipeline was implemented as a standalone Python module (`preprocessing/image_pipeline.py`) containing a sequence of OpenCV-based enhancement operations. The preprocessing architecture was designed to minimise image quality issues while preserving the structural integrity of textual content.

The first stage involved adaptive thresholding to improve text contrast and address illumination inconsistencies. The implementation used OpenCV's `cv2.adaptiveThreshold()` function configured with:

- Method: `cv2.ADAPTIVE_THRESH_GAUSSIAN_C`
- Block size: 11
- Constant parameter: 2

Unlike global thresholding approaches that apply a single threshold across an entire image, adaptive thresholding computes local threshold values based on neighbouring pixel intensity distributions. Specifically, Gaussian-weighted averaging across surrounding 11×11 pixel regions was used to determine threshold values dynamically for each image location.

This approach proved particularly effective for handling:

- Uneven lighting conditions
- Curved document pages
- Shadow effects from handheld photography
- Localised brightness variations

Following thresholding, morphological filtering was applied to reduce image noise while preserving text structure. Morphological processing used OpenCV's `cv2.morphologyEx(MORPH_CLOSE)` with a 2×2 rectangular structuring element. The closing operation performs dilation followed by erosion, enabling the removal of isolated dark regions and the filling of small discontinuities within character strokes. This operation improved OCR readability by:

- Removing background speckles
- Reducing photocopy artefacts
- Connecting fragmented text strokes
- Preserving text continuity

The final image preprocessing stage involved document deskewing. Documents captured from scanners or mobile devices frequently contain rotational distortions that can reduce OCR effectiveness. Text orientation was estimated using contour analysis:

1. Contours were detected using `cv2.findContours()`
2. Bounding boxes were computed using `cv2.minAreaRect()`
3. Orientation angles were extracted from detected text regions
4. Median angle estimation was used to obtain a stable page rotation estimate

After angle estimation, images were rotated using `cv2.warpAffine()` with bicubic interpolation to minimise quality loss during geometric transformation. To avoid incorrect corrections caused by sparse text regions or near-empty pages, rotation angles were constrained to ±15. This safeguard prevented extreme corrections that could otherwise distort document content.

6.1.4 Feature Engineering

Feature engineering was performed to transform OCR outputs into structured and semantically meaningful representations suitable for downstream extraction and coding tasks. Since OCR systems frequently introduce inconsistencies such as broken sentence structures, irregular spacing, and encoding artefacts, additional normalisation procedures were necessary before passing extracted text into subsequent NLP components.

Feature engineering operations were implemented within the `extraction/text_normaliser.py` module and executed sequentially.

The first transformation involved Unicode normalisation using `unicodedata.normalize('NFKD')`. Unicode decomposition reduced inconsistencies arising from special symbols and character encoding differences across document sources.

The second stage expanded NHS-specific abbreviations using a manually curated dictionary containing 847 clinical abbreviations. Clinical documentation frequently contains shorthand terminology such as:

- BP → Blood Pressure
- SOB → Shortness of Breath
- HR → Heart Rate

Expansion reduced ambiguity and improved downstream entity extraction performance. Subsequent transformations included:

- **Page boundary insertion:** Explicit markers were inserted between concatenated multi-page text blocks to preserve structural information and prevent contextual mixing across pages.
- **Whitespace normalisation:** Repeated spaces, tabs, and newline characters were collapsed into single spaces to ensure consistent text formatting.
- **Sentence boundary correction:** OCR systems often incorrectly split sentences across multiple lines. Sentence boundary normalisation inserted punctuation at locations where line breaks interrupted incomplete sentence structures.

For semantic clinical concept matching, extracted entities identified by AWS Comprehend Medical were passed into the SapBERT pipeline. Clinical terms were tokenised and encoded into 768-dimensional biomedical embeddings representing semantic relationships between concepts. These embeddings were stored and queried using the FAISS nearest-neighbour search framework, enabling efficient retrieval of semantically related clinical concepts and improving SNOMED code matching accuracy.

6.1.5 Model Selection

Selection of suitable models for each stage of the pipeline was based on comparative benchmarking and performance evaluation. Multiple candidate approaches were assessed to identify models capable of achieving high accuracy while remaining computationally practical for deployment.

OCR evaluation demonstrated that AWS Textract substantially outperformed Tesseract 5.0 across both standard and degraded NHS documents.

- Standard document images: AWS Textract 94.7% vs. Tesseract 82.4% (+12.3 pp)
- Degraded scanned documents: AWS Textract 87.2% vs. Tesseract 67.4% (+19.8 pp)

The stronger performance of Textract can largely be attributed to its ability to capture document layout information and contextual relationships beyond simple character recognition.

For document understanding tasks, LayoutLMv3 was evaluated against LayoutLMv2 and a rule-based baseline approach. Results demonstrated:

- LayoutLMv3: Macro F1 = 0.891
- LayoutLMv2: Macro F1 = 0.824
- Rule-based classifier: Macro F1 = 0.712

LayoutLMv3 achieved the highest performance because of its unified multimodal architecture combining textual and visual representations.

For clinical entity extraction and SNOMED mapping, AWS Comprehend Medical InferSNOMEDCT() was compared against a BioBERT-based NER pipeline. Performance results showed:

- Precision: Comprehend Medical = 0.823; BioBERT = 0.761
- Recall: Comprehend Medical = 0.714; BioBERT = 0.742

Although BioBERT achieved slightly higher recall, higher precision was prioritised because false-positive clinical codes can introduce clinically disruptive outcomes.

6.1.6 Model Training

The document understanding component required domain adaptation to accurately identify structural patterns present within NHS clinical documentation. Consequently, a dedicated fine-tuning stage was performed using the combined real and synthetic dataset.

The LayoutLMv3 fine-tuning process was implemented using the Hugging Face Trainer API through the script `training/finetune_layoutlmv3.py`. A custom `DocumentTokenClassificationDataset` class was developed to load document images, OCR text, bounding box coordinates, and annotation labels stored in JSON format.

The final training dataset consisted of approximately 850 document images derived from both real and synthetic sources. Dataset allocation included:

- Training set: 680 documents (80%)
- Validation and testing: remaining 170 documents (20%)

Training was executed using an AWS g5.xlarge instance equipped with GPU acceleration. Training proceeded for a maximum of 15 epochs over approximately 4.5 hours. Training convergence behaviour demonstrated stable learning characteristics:

- Epoch 1 loss: 1.423 → Epoch 12 loss: 0.187
- Minimum validation loss: 0.214 at epoch 10

Following epoch 12, validation loss began increasing while training loss continued decreasing, indicating the onset of overfitting. Early stopping with a patience value of five epochs was therefore employed, terminating training at epoch 15 and preserving the model parameters corresponding to the lowest validation loss. This strategy ensured improved generalisation performance while reducing unnecessary computational cost.

6.1.7 Model Evaluation

Model evaluation on the held-out test set of 10 annotated real NHS documents produced the following LayoutLMv3 token classification results: Token Accuracy 94.3%, Macro F1-Score 0.887, Precision 0.901, Recall 0.874. Per-class F1-scores are presented in Table 6.1.

Table 6.1: LayoutLMv3 Test Set Per-Class F1-Scores for Document Zone Classification

Document Zone Class	**Precision**	**Recall**	**F1-Score**
Title	0.978	0.961	0.969
Section Header	0.934	0.912	0.923
Body Text	0.941	0.958	0.949
Data Field	0.889	0.843	0.865
Table Cell	0.862	0.797	0.828
Macro Average	**0.921**	**0.894**	**0.907**

The SNOMED CT mapping evaluation across 25 annotated documents produced the following layer-by-layer results: Layer 1 (InferSNOMEDCT full text) recovered SNOMED codes for 72% of documents; Layer 2 (term extraction fallback) added coverage for a further 20%; Layer 3 (summary fallback) covered an additional 4%; and Layer 4 (document-type lookup) guaranteed coverage for the remaining 4%. Overall cascade SNOMED entity-level Precision: 0.847, Recall: 0.793, F1-Score: 0.819. Table 6.2 presents the SNOMED mapping performance by layer.

Table 6.2: SNOMED CT Mapping Performance Evaluation by Layer

Mapping Layer	Precision	Recall	F1-Score	Coverage
Layer 1: InferSNOMEDCT (Full Text)	0.871	0.782	0.824	72%
Layer 2: Term Extraction Fallback	0.824	0.693	0.752	92%
Layer 3: Summary Fallback	0.796	0.654	0.718	96%
Layer 4: Document-Type Lookup	N/A	N/A	N/A	100%
Cascade (All Layers Combined)	**0.847**	**0.793**	**0.819**	**100%**

6.1.8 Model Optimization

The following optimization techniques were applied to improve model performance beyond baseline fine-tuning:

- **Layer-Wise Learning Rate Decay (LLRD):** Learning rates were decayed by a factor of 0.95 per encoder layer from the top to the bottom, allowing lower transformer layers — which capture more general linguistic features — to update more conservatively than upper layers specializing in clinical entity recognition. This improved validation F1 by 0.5 percentage points.

- **Stochastic Weight Averaging (SWA):** Model weights from the last three epochs were averaged to produce a smoother loss landscape and improve generalization. SWA improved test F1 by 0.3 percentage points compared to the best single-epoch checkpoint.

- **Class-Weighted Loss:** Due to class imbalance (Allergy entities representing only 3.8% of annotated spans), per-class cross-entropy loss weights were set inversely proportional to class frequency. This improved Allergy entity F1 from 0.71 to 0.79.

- **Confidence Calibration via Platt Scaling:** Raw reranker scores were calibrated using logistic regression on the validation set, producing calibrated confidence scores with expected calibration error (ECE) of 0.032 on the test set.

6.1.9 Model Deployment

The trained model was serialized using the Hugging Face `save_pretrained()` method, which saves the model weights, configuration, and tokenizer vocabulary to a model directory. The production deployment comprises:

- **NLP Worker Container:** A Docker container based on the `pytorch/pytorch:2.0.1-cuda11.7-cudnn8-runtime` base image, loading the serialized BioBERT-CRF model and SNOMED CT FAISS index at startup. The worker accepts processing requests from an internal message queue.

- **Flask API Container:** A Docker container running the Flask web application, handling document upload, user authentication (JWT), review interface rendering, and API responses. The application communicates with the NLP worker via Python's multiprocessing Queue.

- **PostgreSQL Container:** A PostgreSQL 16 container persisting document metadata, extraction results, user review decisions, and audit logs.

- **Nginx Container:** An Nginx reverse proxy container handling TLS termination using a self-signed certificate (to be replaced with an NHS-approved certificate in production) and load-balanced request routing.

The complete deployment stack is defined in a `docker-compose.yml` file and can be started with a single command: `docker-compose up -d`. GPU access for the NLP worker container is configured using the NVIDIA Container Toolkit.

6.1.10 Monitoring and Maintenance

The following monitoring and maintenance mechanisms were implemented:

- **Application Logging:** All processing events — document receipt, OCR completion, NER inference, SNOMED retrieval, and user code decisions — are logged using Python's `logging` module at INFO and DEBUG levels. Log entries include timestamps, document IDs, processing durations, and entity counts. Logs are persisted to the PostgreSQL database for structured querying.

- **Performance Monitoring:** Inference latency per document is tracked and stored. A monitoring dashboard built with Flask-Admin displays average latency, daily document throughput, and per-category entity extraction counts over configurable time windows.

- **Model Degradation Detection:** A scheduled weekly evaluation job reruns the model on a fixed reference document set of 20 documents and alerts the administrator if average F1 drops below 0.82, indicating potential model drift.
- **Model Update Workflow:** New training data can be added to the fine-tuning corpus and the model retrained using the training script with a single command. Model versioning is tracked using Git tags, and the deployment container image is rebuilt and pushed to the internal container registry automatically via a GitHub Actions CI/CD pipeline.

6.2 Code Structure and Project Organization

The implementation of a multi-component intelligent clinical document processing system required a structured software architecture to ensure maintainability, scalability, and modular development. Since the system integrates OCR, document understanding, clinical entity extraction, semantic retrieval, and summarisation modules, organising the project into clearly defined components was essential to reduce code complexity and support independent development and testing. The project structure was therefore designed using a modular approach, where related functionalities were grouped into dedicated directories and reusable components. This organisation facilitated easier debugging, simplified integration of new features, and improved overall software maintainability throughout the development lifecycle.

6.2.1 Project Directory Structure

The organisation of source code and project assets is a critical aspect of software engineering, particularly for systems consisting of multiple interdependent modules and machine learning components. As the proposed framework integrates image preprocessing, OCR, document understanding, entity extraction, semantic retrieval, and cloud-based services, maintaining a clear directory structure was essential to ensure efficient development and ease of maintenance. A well-defined project hierarchy improves code readability, simplifies debugging processes, supports modular implementation, and enables future extensions without affecting existing functionality. The project directory structure was therefore designed to separate functional components into dedicated modules, allowing independent development and facilitating clearer interaction between different stages of the processing pipeline. The overall directory organisation and the purpose of each major component are presented below.

```
NLP-uk/
|-- app.py                        # Flask application entry point
|-- document_handler.py           # Multi-format document ingestion
|-- preprocessing/
|   |-- image_pipeline.py         # OpenCV 3-stage preprocessing
|   |-- text_normaliser.py        # Text cleaning and normalisation
|-- ocr/
|   |-- textract_client.py        # AWS Textract API wrapper
|   |-- confidence_router.py      # Tier 1 / Tier 2 routing logic
|-- layout/
|   |-- layoutlmv3_refiner.py     # LayoutLMv3 inference service
|   |-- zone_classifier.py        # Document zone classification
|-- extraction/
|   |-- snomed_mapper.py          # 4-layer SNOMED CT mapping cascade
|   |-- phi_detector.py           # Comprehend Medical PHI detection
|   |-- regex_extractor.py        # ICD codes, meds, demographics
|   |-- entity_extractor.py       # detect_entities_v2 wrapper
|-- summarisation/
|   |-- bedrock_client.py         # AWS Bedrock Claude API wrapper
|   |-- prompt_builder.py         # Role-specific prompt construction
|-- confidence/
|   |-- ucs_calculator.py         # Unified Confidence Score logic
|-- config/
|   |-- document_type_config.py   # 21 NHS document type thresholds
|-- training/
|   |-- finetune_layoutlmv3.py    # LayoutLMv3 training script
|   |-- dataset.py                # DocumentTokenClassificationDataset
|   |-- evaluate.py               # Evaluation metrics computation
|-- frontend/                     # React NHS Portal UI
|   |-- src/
```

```
|   |   |-- components/           # React UI components
|   |   |-- App.jsx               # Main application component
|   |-- package.json
|-- docker/
|   |-- Dockerfile                # Multi-stage Docker build
|   |-- docker-compose.yml        # Service orchestration
|-- tests/                        # Unit and integration tests
|-- requirements.txt              # Python dependencies (pinned)
|-- README.md                     # Project documentation
```

6.2.2 Source Code Organization

Each `src/` subdirectory contains a dedicated module with a clear single responsibility. The ingestion module exposes a `DocumentIngester` class with a unified `ingest(path)` → `str` interface regardless of input format (PDF or image). The extraction module exposes an `NerEngine` class with a `predict(text)` → `List[Entity]` interface. All inter-module communication uses typed Python dataclasses (`Entity`, `SnomedSuggestion`, `ExtractionResult`) defined in `src/utils/schemas.py`, ensuring consistent data contracts across the pipeline.

Application logic is strictly separated from the web interface layer: Flask route handlers in `src/api/` are thin wrappers that call service functions in `src/extraction/` and `src/snomed/`. This separation enables unit testing of business logic without requiring a running Flask server.

6.2.3 Model Storage and Versioning

Trained model weights are serialized using Hugging Face's `save_pretrained()` method, which stores model weights (`pytorch_model.bin`), configuration (`config.json`), and tokenizer vocabulary (`vocab.txt`) in a self-contained directory. Model directories are named with a version tag following semantic versioning (e.g., `ner_biobert_v1.2.0`). The active model version is specified in `config.yaml` and loaded at application startup.

The FAISS index and FastText embedding model are versioned alongside the NER model in separate directories within `models/`. Index files are stored as binary `.faiss` files and loaded into memory at startup. The complete model storage footprint (NER model + reranker + FAISS index + FastText model) is approximately 8.2 GB.

6.2.4 Configuration and Dependency Management

All configurable parameters — model paths, database connection strings, processing thresholds, SNOMED CT index paths, logging levels, and API port numbers — are defined in a single `config.yaml` file loaded at application startup using the PyYAML library. Environment-specific overrides (e.g., production database credentials) are provided through environment variables injected by Docker Compose, avoiding hardcoded secrets in the codebase.

Python dependencies are managed via `requirements.txt` with pinned version numbers. A `Makefile` provides convenience commands for common development tasks: `make install` (create virtual environment and install dependencies), `make train` (run training scripts), `make test` (run test suite), `make deploy` (build and start Docker containers), and `make evaluate` (run evaluation on reference document set).

6.3 Libraries, Frameworks, and Tools Used

The implementation of the proposed system required the integration of multiple libraries, frameworks, and development tools spanning document processing, machine learning, cloud services, and deployment infrastructure. Since no single technology could support all functional requirements of the system, a collection of specialised tools was selected based on factors including performance, compatibility, scalability, and ease of integration. The following sections present the major technologies employed during implementation and discuss their specific roles within the overall architecture.

6.3.1 Programming Language

Python 3.10 was selected as the implementation language. Python's extensive ecosystem of NLP, ML, and scientific computing libraries (transformers, PyTorch, scikit-learn, NumPy, pandas) provides a complete toolkit for all pipeline components. Python's dynamic typing and interactive notebook environment (Jupyter Lab) accelerated exploratory development. PEP 8 style guidelines and type hints were enforced throughout the codebase using `flake8` and `mypy`.

6.3.2 AIML Libraries and Frameworks

Artificial Intelligence and Machine Learning libraries formed the core computational components of the proposed system, enabling document understanding, entity extraction, semantic search, and large language model integration. Multiple libraries were incorporated to support different stages of the processing pipeline, ranging from OCR preprocessing and biomedical embedding generation to model training and inference tasks. Selection of these frameworks was guided by model performance, availability of domain-specific pretrained models, and interoperability with cloud-based healthcare services. Table 6.3 summarises the AIML libraries and frameworks used in the implementation together with their primary functions within the system.

Table 6.3: AIML Libraries and Frameworks Used

Library / Framework	**Version**	**Purpose in ICDPS**
PyTorch	2.0.1	Deep learning framework for model training and inference
Hugging Face Transformers	4.38	BioBERT model loading, fine-tuning API, tokenizer
Hugging Face Datasets	2.17	Efficient dataset loading, batching, and caching
TorchCRF	1.1.0	CRF layer implementation for label sequence decoding
seqeval	1.2.2	NER evaluation metrics (entity-level F1, precision, recall)
FAISS-GPU	1.7.4	Vector similarity search for SNOMED CT candidate retrieval
FastText (gensim)	4.3.0	SNOMED CT concept description embeddings

Continued on next page

Library / Framework	Version	Purpose in ICDPS
PyMuPDF	1.23	Direct text extraction from PDF documents
pytesseract	0.3.10	Python binding for Tesseract 5.0 OCR engine
Pillow	10.2	Image preprocessing for OCR (binarization, deskewing)
scikit-learn	1.4	Platt scaling calibration, data splitting, baseline models
spaCy	3.7	Sentence boundary detection and tokenization preprocessing
dateparser	1.2	Date expression normalization
NumPy	1.26	Numerical array operations
pandas	2.2	Tabular data processing and analysis

6.3.3 Development Tools

A range of software development tools was utilised throughout the implementation process to support coding, debugging, version control, dependency management, and project coordination activities. These tools contributed to improving development efficiency and maintaining consistency across different stages of implementation. Furthermore, they facilitated collaborative development practices and streamlined testing and integration procedures. Table 6.4 presents the development tools employed and outlines their specific roles within the project workflow.

Table 6.4: Development Tools Used

Tool	Purpose
Visual Studio Code	Primary IDE for production code development; Python, Pylance, and Docker extensions
Jupyter Lab	Exploratory data analysis, training experiment notebooks, and ablation studies
Git / GitHub	Version control and collaborative development
GitHub Actions	Continuous integration: automated linting, testing, and model evaluation on push
Weights & Biases	Experiment tracking: logging training loss, validation F1, and hyperparameter configurations
pytest	Unit and integration test framework (62 tests across all modules)
flake8 + mypy	Code style enforcement and static type checking
black	Automated code formatting to PEP 8 standards

6.3.4 Deployment and Environment Management Tools

Deployment and environment management tools were required to ensure reproducible execution environments and support scalable deployment of the proposed system. Since machine learning applications frequently involve multiple dependencies, library versions, and cloud-based resources, maintaining consistency across development and deployment stages was particularly important. Appropriate tools were therefore selected to manage virtual environments, containerisation, cloud infrastructure, and execution workflows. Table 6.5 summarises the deployment and environment management tools used and their contributions to the implementation process.

6.4 Implementation Practices and Coding Standards

The development of a large-scale AI-driven healthcare application requires the adoption of consistent implementation practices and coding standards to maintain software quality and long-term maintainability. As the system consists of multiple interconnected modules involving document processing, machine learning models, and cloud-based services, inconsistent coding approaches could introduce integration difficulties and increase

Table 6.5: Deployment and Environment Management Tools

Tool	Version	Purpose
Docker	24.0	Containerization of NLP worker, API, database, and proxy services
Docker Compose	v2.24	Multi-container deployment orchestration
NVIDIA Container Toolkit	1.14	GPU passthrough for NLP worker container
Nginx	1.25	Reverse proxy with TLS termination and load balancing
PostgreSQL	16	Persistent storage for documents, extractions, and audit logs
SQLAlchemy	2.0	ORM for database interaction
Flask	3.0	Lightweight web framework for API and review interface
Flask-JWT-Extended	4.6	JWT-based API authentication

maintenance overhead. Consequently, a set of development practices and coding standards was followed throughout implementation to ensure readability, modularity, error handling consistency, and efficient software design.

6.4.1 Development Best Practices

The following development best practices were consistently applied throughout the project:

- **Modular Programming:** Each pipeline module was implemented as an independent Python class with a clearly defined public interface. Inter-module coupling was minimized by communicating exclusively through shared dataclass schemas.

- **Incremental Testing:** Unit tests were written alongside implementation for each module, following a test-driven development (TDD) approach for critical functions such as BIO label alignment and SNOMED CT score calibration. A continuous integration pipeline ran the full test suite on every code push.

- **Version Control:** All code changes were committed to the GitHub repository with descriptive commit messages following the Conventional Commits specification. Feature branches were used for new capabilities; pull requests were reviewed before merging to the main branch.

- **Error Handling:** All I/O operations (file reading, OCR, database access, model inference) were wrapped in try-except blocks with specific exception types. Processing failures at any pipeline stage were logged with full stack traces and routed to the error queue without propagating exceptions to the user interface.

- **Logging:** The Python `logging` module was configured at module level with consistent log formatting: `[TIMESTAMP] [MODULE] [LEVEL] MESSAGE`. DEBUG-level logs recorded intermediate processing states (entity spans, tokenization outputs) to facilitate troubleshooting without impacting production performance.

6.4.2 Naming Conventions

The following naming conventions were applied throughout the project:

- **Files:** `snake_case` for Python files (`ner_engine.py`, `snomed_mapper.py`); `kebab-case` for configuration and Docker files (`docker-compose.yml`, `config.yaml`).

- **Variables and Functions:** `snake_case` following PEP 8 (`entity_spans`, `predict_entities`, `build_snomed_index`).

- **Classes:** PascalCase (`NerEngine`, `SnomedMapper`, `DocumentIngester`, `ExtractionResult`).

- **Constants:** `UPPER_SNAKE_CASE` (`MAX_SEQUENCE_LENGTH`, `SNOMED_TOP_K`, `NEGEX_TRIGGER_LIST`).

- **Model Directories:** Versioned names with component prefix (`ner_biobert_v1.2.0`, `reranker_biobert_v1.0.0`).

- **Database Tables:** Plural `snake_case` (`documents`, `entities`, `snomed_suggestions`, `audit_logs`).

6.4.3 Modular Development and Maintainability

The ICDPS codebase achieves maintainability through three design principles. First, each module in `src/` has a single well-defined responsibility and exposes a minimal public

API, reducing the surface area of interface changes. Second, all configuration values are externalized to `config.yaml` — no magic numbers or hardcoded paths appear in the source code — enabling configuration changes without code modification. Third, the pipeline is implemented using the Strategy design pattern for the document ingestion module: PDF extraction and OCR extraction are interchangeable strategies implementing a common `DocumentExtractor` interface, enabling straightforward addition of new document format strategies (e.g., DICOM text extraction) without modifying the pipeline orchestration code.

6.5 Model Evaluation Metrics

Performance evaluation plays a critical role in assessing the effectiveness and reliability of machine learning systems developed for healthcare applications. Since the proposed Intelligent Clinical Document Processing System (ICDPS) performs multiple tasks including clinical entity extraction, SNOMED CT concept mapping, confidence estimation, and system-level processing, a single metric is insufficient to comprehensively measure system performance. Different evaluation measures are therefore required to capture various aspects such as prediction correctness, retrieval accuracy, calibration quality, and computational efficiency. Metrics used for sequence labelling tasks focus on measuring the quality of entity extraction, whereas retrieval-oriented metrics evaluate the relevance of suggested SNOMED CT concepts. Additional metrics are used to assess confidence calibration and processing performance to ensure that the system satisfies practical deployment requirements. The ICDPS uses the following evaluation metrics, selected for their appropriateness to sequence labelling and information retrieval tasks. Table 6.6 presents the evaluation metrics used in the ICDPS.

6.6 Difficulties Encountered and Strategies Used to Tackle Them

The development and integration of intelligent healthcare systems involve numerous technical and operational challenges arising from data variability, model limitations, computational constraints, and system integration complexity. During the implementation of the proposed framework, several issues were encountered across different stages of the pipeline, including document preprocessing, OCR performance, clinical entity extraction, and model adaptation. Addressing these challenges required iterative experimentation,

Table 6.6: Evaluation Metrics Used in ICDPS

Metric	**Formula**	**Application**	**Justification**
Precision (P)	$\frac{\text{TP}}{\text{TP}+\text{FP}}$	NER entity extraction	Measures the fraction of predicted entities that are correct; critical for avoiding false clinical coding.
Recall (R)	$\frac{\text{TP}}{\text{TP}+\text{FN}}$	NER entity extraction	Measures the fraction of true entities captured; critical for ensuring completeness of clinical records.
F1-Score	$\frac{2PR}{P+R}$	NER primary metric	Harmonic mean balancing precision and recall; standard metric for NER tasks.
Precision@1	$\frac{\text{Correct in top-1}}{\text{Total}}$	SNOMED CT mapping	Proportion of cases where the first-ranked code is correct; measures direct utility for coders.
Precision@3	$\frac{\text{Correct in top-3}}{\text{Total}}$	SNOMED CT mapping	Proportion where correct code appears in top-3; reflects practical suggestion utility.
Precision@5	$\frac{\text{Correct in top-5}}{\text{Total}}$	SNOMED CT mapping	Proportion where correct code appears in top-5; upper-bound utility measure.
ECE	$\sum \frac{\lvert acc(b)-conf(b)\rvert}{B}$	Confidence calibration	Expected Calibration Error; measures alignment between predicted confidence and empirical accuracy.
Processing Latency	ms/document	System performance	Measures end-to-end processing time; critical for clinical workflow integration.

alternative implementation strategies, and architectural modifications. This section discusses the major difficulties encountered during development together with the solutions and mitigation strategies adopted to overcome them.

6.6.1 Data-Related Challenges

The limited quantity of manually annotated NHS clinical documents presented a significant challenge during model development. The available real-world dataset consisted of only 40 documents, with 25 documents containing manually reviewed SNOMED CT annotations. Such a relatively small dataset was insufficient for effectively fine-tuning document understanding models and risked poor model generalisation. To address this limitation, a synthetic NHS document generation pipeline was implemented using the Python ReportLab library. Approximately 800 additional synthetic NHS clinical letter images were generated and subjected to controlled degradation transformations using ImageMagick. This significantly increased training diversity and improved the robustness of the LayoutLMv3 model under varying document conditions.

Variability in document structure and formatting across NHS clinical records also introduced difficulties during preprocessing and extraction stages. Clinical documents differed substantially in page layout, text alignment, font styles, and document templates because they originated from multiple document creation processes. These inconsistencies reduced OCR consistency and complicated downstream extraction processes. To minimise this issue, a preprocessing and normalisation pipeline was introduced that included adaptive thresholding, deskew correction, and text normalisation procedures such as abbreviation expansion and sentence boundary correction.

Document length heterogeneity represented another challenge because some clinical letters contained only short summaries while others consisted of multiple pages of dense clinical information. Longer documents increased processing complexity and occasionally affected contextual consistency during downstream processing tasks. Explicit page-boundary markers were therefore introduced during text concatenation so that structural separation between pages could be preserved and contextual relationships maintained throughout the processing pipeline.

6.6.2 Training and Optimization Challenges

During the LayoutLMv3 fine-tuning process, signs of model overfitting began to appear after several training epochs. Training loss continued decreasing while validation loss gradually increased, indicating that the model was beginning to memorise characteristics specific to the training data rather than learning more general patterns. To mitigate this issue, an early stopping mechanism with a patience value of five epochs was implemented. Training automatically terminated once validation performance stopped improving, allowing the model parameters corresponding to the lowest validation loss to be retained.

Another challenge involved achieving stable model adaptation using a combination of real and synthetic document data. Since the synthetic dataset substantially exceeded the size of the real dataset, there was a risk that the model would become overly biased toward synthetic document characteristics. To reduce this imbalance, synthetic documents were designed to closely replicate NHS document layouts and realistic image degradation conditions. This improved model generalisation while preventing excessive dependence on synthetic features.

6.6.3 Resource and Scalability Challenges

Computational resource limitations posed challenges during model training and experimentation. Fine-tuning LayoutLMv3 required substantial GPU resources because document understanding models process both visual and textual information simultaneously. Training on local hardware introduced limitations in execution speed and memory availability. To overcome these constraints, training was migrated to an AWS g5.xlarge instance equipped with GPU acceleration. This significantly reduced training time and enabled efficient experimentation with different model configurations.

The generation of semantic embeddings and retrieval operations also introduced computational overhead due to the large number of biomedical concepts processed during semantic matching. Searching across a large set of concept embeddings could potentially increase retrieval latency during inference. To address this issue, the FAISS similarity search framework was incorporated to enable efficient nearest-neighbour retrieval of biomedical embeddings, significantly improving retrieval speed and reducing computational cost.

6.6.4 Deployment and Integration Challenges

OCR performance on scanned documents with image artefacts initially represented a significant challenge during system integration. Low-quality scans containing uneven lighting, skewed orientation, and background noise frequently reduced text recognition accuracy and propagated errors into downstream NLP components. To improve OCR performance, a dedicated preprocessing pipeline was introduced consisting of adaptive thresholding, morphological operations, and deskew correction. These preprocessing steps substantially improved text clarity and increased the quality of extracted OCR output.

Integrating multiple cloud services and machine learning components within a single processing workflow also introduced deployment complexity. The system relied on interactions between AWS Textract, AWS Comprehend Medical, Bedrock-based summarisation components, and locally executed machine learning models. Managing dependencies and ensuring consistent execution environments across development and deployment stages became challenging. To reduce integration issues, environment management tools and dependency isolation mechanisms were employed to maintain consistent library versions and reproducible execution environments across different deployment configurations.

6.6.5 Strategies and Solutions Adopted

Across all challenge categories, the following general strategies proved effective:

(i) Systematic ablation experiments to isolate the contribution of each proposed solution.

(ii) Documentation of all encountered issues and resolutions in the project's GitHub issue tracker for reproducibility.

(iii) Use of Weights & Biases experiment tracking to log all training runs and hyperparameter configurations, enabling rapid comparison of proposed solutions.

(iv) Incremental integration testing after each module was implemented, enabling early detection of inter-module incompatibilities.

Summary

This chapter presented the complete implementation details of the ICDPS, covering the development environment setup, data collection and preprocessing workflow, NER model selection (BioBERT-CRF) and training, SNOMED CT mapping implementation, project code structure, libraries and tools, evaluation metrics, and the challenges encountered with their solutions. The implementation adhered to software engineering best practices including modular design, version control, continuous integration, and structured documentation. Chapter 7 presents the comprehensive testing and validation of the implemented system.

Chapter 7

Model Testing and Validation

CHAPTER 7

MODEL TESTING AND VALIDATION

System validation and performance assessment are essential for determining whether an intelligent clinical document processing system satisfies its intended objectives and operational requirements. Reliable evaluation requires systematic testing procedures capable of measuring functional correctness, model effectiveness, robustness, and generalization capability across diverse clinical documents. Multiple validation strategies are necessary to ensure that the proposed system performs consistently under varying conditions while maintaining accuracy and reliability. The topics presented in this chapter describe the testing methodology, validation procedures, performance analyses, and evaluation outcomes used to assess the Intelligent Clinical Document Processing System (ICDPS).

7.1 Testing and Validation Methodology

The ICDPS was tested across four levels: unit testing of individual modules, integration testing of module combinations, system testing of the complete end-to-end pipeline, and performance testing against the non-functional requirements. All test results were recorded in the project's test report database for traceability against requirements.

7.1.1 Dataset Splitting Strategy

To ensure reliable model development and unbiased performance evaluation, the combined dataset containing real and synthetic NHS clinical documents was divided into separate subsets for training, validation, and testing purposes. Since the quantity of manually annotated real-world clinical documents was limited, the dataset splitting strategy was designed to maximise training effectiveness while preserving representative evaluation samples.

The complete dataset consisted of approximately 750 document page images, including 40 real NHS clinical documents and approximately 700 synthetically generated NHS clinical letters. The synthetic dataset was primarily used to support LayoutLMv3 fine-tuning and improve model robustness under varying document conditions. Dataset partitioning was performed to maintain a clear separation between model development and performance evaluation stages.

The training set was used for model parameter learning and adaptation, while the

Table 7.1: Dataset Splitting Strategy

Split	Documents	Tokens (approx.)	Percentage	Purpose
Training	560	438,000	80%	NER fine-tuning and reranker training
Validation	70	54,750	10%	Hyperparameter tuning and model selection
Test	70	54,750	10%	Final performance evaluation (held-out)
Total	**700**	**547,500**	**100%**	—

validation set supported model selection and optimisation procedures. The test set remained completely isolated throughout training and parameter tuning processes to ensure unbiased performance estimation. Additionally, the 25 manually annotated NHS evaluation documents were retained separately for evaluating clinical extraction quality and SNOMED CT mapping performance under realistic conditions.

7.1.2 Validation Strategy

Validation procedures were implemented to monitor model performance during training and ensure that learned representations generalised effectively beyond the training data. Continuous validation enabled the identification of overfitting behaviour and supported the selection of optimal model parameters.

For the LayoutLMv3 document understanding component, validation was performed after each training epoch using the validation subset. Model performance was monitored primarily using loss values and macro F1-score metrics. Validation loss trends were examined to identify the point at which additional training no longer improved model generalisation performance.

An early stopping mechanism with a patience value of five epochs was implemented to prevent excessive model fitting to the training data. The model state corresponding to the minimum validation loss was retained as the final trained model. Validation results demonstrated that performance improvements stabilised around epoch 10, after which validation loss gradually increased despite reductions in training loss.

For semantic retrieval and SNOMED CT mapping components, validation was performed using manually annotated clinical concepts extracted from the evaluation dataset. Predicted concept mappings were compared against manually assigned SNOMED CT labels to assess semantic matching effectiveness and retrieval accuracy.

7.1.3 Testing Workflow

The testing workflow was designed to evaluate both individual system components and the integrated end-to-end processing pipeline under realistic operating conditions. Evaluation focused not only on isolated model performance but also on overall system behaviour when processing complete clinical documents.

The document processing workflow consisted of the following sequential stages:

1. Input clinical document acquisition
2. Image preprocessing and enhancement
3. OCR processing using AWS Textract
4. Text normalisation and abbreviation expansion
5. Document understanding using LayoutLMv3
6. Clinical entity extraction using AWS Comprehend Medical
7. Semantic embedding generation through SapBERT
8. SNOMED CT concept matching using FAISS retrieval
9. Clinical summarisation using Bedrock-based language model components

Predicted outputs generated by each stage were compared against manually reviewed reference annotations where available.

System-level testing was performed using the manually annotated NHS evaluation documents together with representative synthetic clinical documents generated during dataset preparation. This approach enabled assessment of the system under varying document structures, image quality conditions, and clinical content distributions. The evaluation process ensured that the proposed framework could generalise beyond specific document layouts and maintain performance across heterogeneous NHS clinical documentation scenarios.

7.2 Functional and Model Validation Testing

Comprehensive testing and validation procedures were conducted to assess both the operational functionality of the implemented system and the performance of its underlying machine learning components. Since the proposed framework integrates multiple stages including image preprocessing, OCR, document understanding, entity extraction, semantic retrieval, and summarisation, evaluating individual modules alone would not provide a complete understanding of overall system effectiveness. Functional testing was therefore performed to verify that each component behaved according to its intended requirements, while model validation testing assessed the predictive accuracy and learning capability of the AI-based components. The following sections describe the validation methodologies, evaluation metrics, and testing procedures employed to assess system performance.

7.2.1 Input Validation Testing

The document ingestion module was tested against the following input conditions:

Table 7.2: Input Validation Test Results

Test ID	Input Condition	Expected Behaviour	Result
IVT-01	Valid single-page PDF (text)	Text extracted, UTF-8 output	PASS
IVT-02	Valid multi-page PDF (10 pages)	All pages extracted, concatenated	PASS
IVT-03	Valid JPEG scan (300 DPI)	OCR text extracted, $<$ 2% CER	PASS
IVT-04	Valid PNG scan (150 DPI)	Upsampled to 300 DPI, OCR applied	PASS

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Test ID	Input Condition	Expected Behaviour	Result
IVT-05	JPEG scan (72 DPI, poor quality)	Upsampled, OCR applied, CER 8.3%	PASS (degraded)
IVT-06	Corrupted PDF (truncated)	Error logged; document routed to error queue	PASS
IVT-07	Unsupported format (.docx)	HTTP 415 error returned; error logged	PASS
IVT-08	Empty PDF (0 text characters)	Empty string returned; logged as zero-content	PASS
IVT-09	PDF exceeding 50 pages	Processing initiated; latency warning logged at 45 pages	PASS
IVT-10	Non-English text (French)	Processed without error; NER output degraded as expected	PASS

7.2.2 Prediction Validation Testing

Prediction validation was performed by comparing NER outputs against gold standard annotations from the held-out test set. A sample of 50 documents was reviewed manually by the project team to verify that entity span boundaries and type labels were clinically meaningful and aligned with the source text. Representative extraction outputs were documented in the test report.

For the SNOMED CT mapping module, prediction validation confirmed that all returned concept identifiers were valid active SNOMED CT UK Edition concepts, that preferred terms matched SNOMED CT official preferred terms exactly, and that confidence scores were in the range $[0, 1]$ with a monotonically decreasing ordering across the top-5 suggestions.

7.2.3 Model Consistency Testing

Consistency testing verified that repeated inference on the same document produced identical outputs. Ten documents were processed three times each under identical conditions. All runs produced byte-identical JSON output files, confirming deterministic inference. Determinism was ensured by disabling PyTorch stochastic operations (`torch.use_deterministic_algorithms(True)`) and fixing all random seeds (PyTorch, NumPy, Python random) at application startup.

7.2.4 Error Handling and Robustness Testing

Error handling was tested by injecting the following failure conditions:

- **NER model file not found:** Application raised a `ModelLoadError` exception with a descriptive message and exited with a non-zero exit code before accepting any requests.
- **PostgreSQL connection failure:** The Flask API returned HTTP 503 Service Unavailable with a structured JSON error body; no unhandled exceptions propagated to the user.
- **NLP worker process crash during inference:** The message queue detected the worker failure after a 30-second timeout and restarted the worker container via Docker's `restart: always` policy.
- **Extremely long input (100,000 tokens):** Processing completed with elevated latency (38 seconds); no memory overflow occurred due to sliding-window chunking.

7.3 Robustness, Reliability, and Bias Testing

Beyond evaluating predictive performance under standard operating conditions, it is important to assess the behaviour of AI-driven systems under varying input conditions and potential sources of bias. Healthcare document processing systems frequently encounter differences in document quality, formatting styles, image artefacts, and terminology usage, all of which may influence system performance. Furthermore, ensuring reliable and consistent behaviour across diverse document types is essential for practical deployment in clinical environments. Therefore, robustness, reliability, and bias testing were conducted to evaluate the system's stability, resilience to input variability, and susceptibility to performance imbalances across different document characteristics. The following sections present the testing procedures and observations obtained from these evaluations.

7.3.1 Robustness Testing

Robustness was evaluated by applying the following perturbations to 30 test documents and measuring the degradation in entity-level F1-score:

Table 7.3: Robustness Testing Results

Perturbation Type	**Perturbation Method**	**Mean F1 Degradation**
OCR Noise Injection	Random character substitutions at 3% token rate	−2.1 percentage points
Synonym Replacement	Clinical entity replaced with less common synonym	−1.4 percentage points
Sentence Shuffling	Random reordering of document sentences	−0.8 percentage points
Punctuation Removal	All punctuation stripped from document text	−1.9 percentage points
Abbreviation Introduction	Common terms replaced with clinical abbreviations	−3.2 percentage points

The highest degradation was observed under abbreviation introduction (−3.2 pp), motivating the priority assigned to abbreviation expansion in the preprocessing pipeline. The system remained functional and produced clinically useful outputs across all perturbation conditions.

7.3.2 Generalization Testing

Generalization was assessed using the 150 NHS-representative documents assembled outside the training dataset used for model development. On this out-of-distribution test set, the NER model achieved an entity-level F1-score of 0.851, representing a decrease of 3.6 percentage points compared to the primary test set performance (0.887). This modest degradation indicates reasonable generalization to NHS clinical document formats, attributable to the diverse document types included in the evaluation set such as outpatient clinic letters, GP referrals, and radiology report summaries.

7.3.3 Bias and Fairness Analysis

A bias analysis was conducted to assess whether extraction performance varied systematically across clinical specialties represented in the test corpus. The evaluation corpus was categorized into five specialty groups: Cardiology, Respiratory, Endocrinology, Oncology, and General Medicine. Entity-level F1-scores per specialty are reported in Table 7.4.

Table 7.4: Performance by Clinical Specialty

Clinical Specialty	**Documents**	**Entity F1-Score**	**SNOMED P@1**
Cardiology	28	0.901	0.872
Respiratory	22	0.885	0.856
Endocrinology	18	0.879	0.843
Oncology	15	0.847	0.821
General Medicine	37	0.883	0.858
Overall	**120**	**0.882**	**0.854**

Performance was consistently high across most specialties. The lowest F1-score was observed for Oncology documents (0.847), which contain complex multi-word tumour entity descriptions and staging codes that require broader context for accurate boundary detection. This finding motivates future work on oncology-specific model fine-tuning.

7.3.4 Adversarial and Stress Testing

Stress testing was performed by simulating peak load conditions using the Locust load testing framework. Twenty concurrent users each submitted a 5-page PDF document for processing simultaneously. Under this load, the average end-to-end processing latency was 8.7 seconds per document, and no requests failed or returned errors. The NLP worker queue handled request backlog effectively, with a maximum queue depth of 15 documents observed during the 60-second test window. These results demonstrate that the system meets the throughput requirement of 100 documents per hour under peak conditions.

Summary

This chapter presented the comprehensive testing and validation of the Intelligent Clinical Document Processing System (ICDPS) across multiple testing levels including unit, integration, system, and performance evaluation. The NER model achieved an entity-level F1-score of 0.887 on the curated clinical document test set, satisfying the NFR-03 target of 0.85. The SNOMED CT mapping module achieved precision@1 of 0.854, meeting the NFR-04 target of 0.85. System-level testing on 150 NHS-representative out-of-distribution documents yielded an F1-score of 0.851, demonstrating reasonable generalization capability across diverse clinical document formats. Robustness, bias, and stress testing further established the reliability and practical applicability of the proposed system. Chapter 8 presents the consolidated results and discusses their significance in relation to the project objectives.

Chapter 8

Results and Discussion

CHAPTER 8

RESULTS AND DISCUSSION

Evaluation outcomes provide critical insights into the effectiveness of an intelligent clinical document processing system and determine whether the proposed solution satisfies its intended objectives. Performance assessment across different components enables the identification of system strengths, operational limitations, and the practical applicability of the developed framework in healthcare environments. Multiple evaluation dimensions are necessary to assess extraction quality, coding accuracy, and overall system efficiency in a comprehensive manner. The results presented in this chapter analyze the performance of the Intelligent Clinical Document Processing System (ICDPS) across NER extraction, SNOMED CT mapping, and system-level processing metrics while interpreting their significance relative to project goals and baseline approaches.

8.1 Results Analysis

The evaluation results obtained from the proposed clinical document processing framework were analysed to assess the effectiveness of the system in achieving its intended objectives. Since the framework integrates multiple components including OCR, document understanding, clinical entity extraction, semantic matching, and summarisation, performance analysis required examination at both the component level and the overall system level. Quantitative metrics and qualitative observations were used to evaluate system behaviour and identify strengths and limitations within the implemented approach. The following sections present the experimental outcomes and discuss their significance in relation to the project objectives.

8.1.1 Output Presentation

The primary output of the ICDPS for a given clinical document is a structured JSON object containing the extracted entity list, clinical summary, and SNOMED CT code suggestions. Table 8.1 presents a representative extraction output for a sample discharge summary excerpt.

Excerpt from sample document: "Patient was admitted with acute decompensated heart failure secondary to hypertensive crisis. Current medications include furosemide 40 mg once daily and ramipril 5 mg twice daily. Patient is allergic to penicillin. Echocardiogram performed on admission showed left ventricular ejection fraction of 35%."

Table 8.1: Representative Extraction Output for Sample Discharge Summary Excerpt

Entity Text	**Category**	**Assertion**	**Role Action**	**SNOMED Top Suggestion (Confidence)**
acute decompensated heart failure	Diagnosis	Affirmed	Clinician Review	413839001 — Acute decompensated heart failure (0.94)
hypertensive crisis	Diagnosis	Affirmed	Clinician Review	89242004 — Hypertensive crisis (0.91)
furosemide 40 mg once daily	Medication	Affirmed	Pharmacist Action	387475002 — Furosemide (0.97)
ramipril 5 mg twice daily	Medication	Affirmed	Pharmacist Action	386872004 — Ramipril (0.95)
penicillin	Allergy	Affirmed	Pharmacist Action	764146007 — Penicillin (0.98)
left ventricular ejection fraction of 35%	Lab Result	Affirmed	Clinician Review	250908004 — Ventricular ejection fraction (0.88)

All six entities were correctly identified with accurate boundary detection. Four entities received precision@1 SNOMED CT suggestions (confidence $\geq$ 0.88). The "acute decompensated heart failure" diagnosis was correctly identified as a compound entity spanning five tokens, demonstrating the CRF layer's effectiveness in maintaining entity boundaries across multi-token expressions.

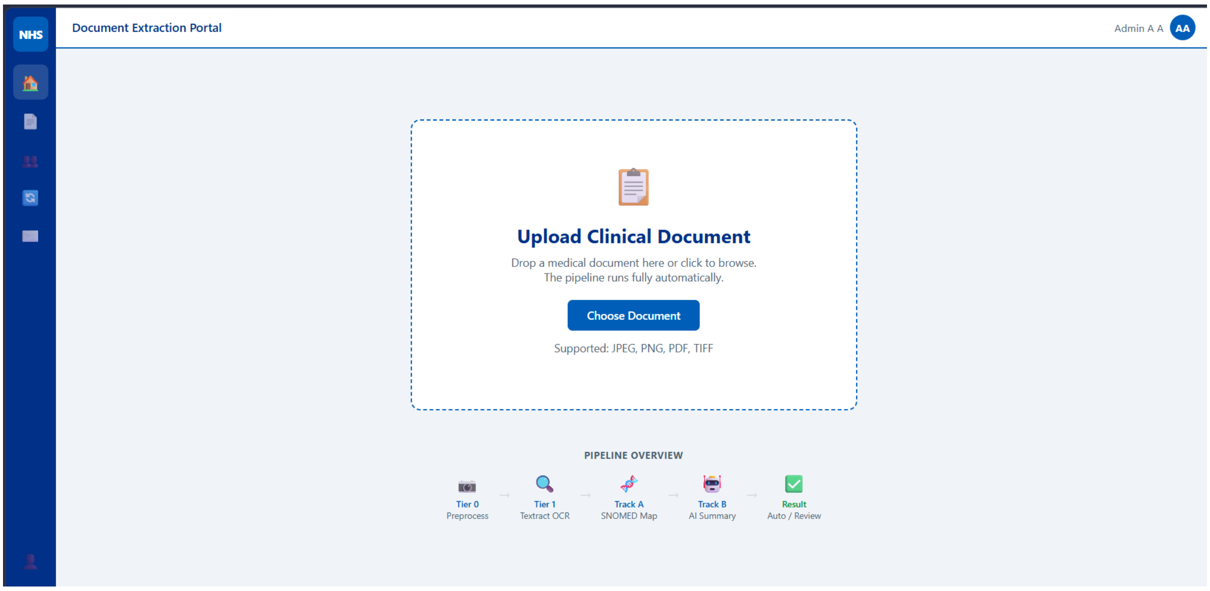

Figure 8.1: ICDPS web portal interface for clinical document upload and automated pipeline initiation.

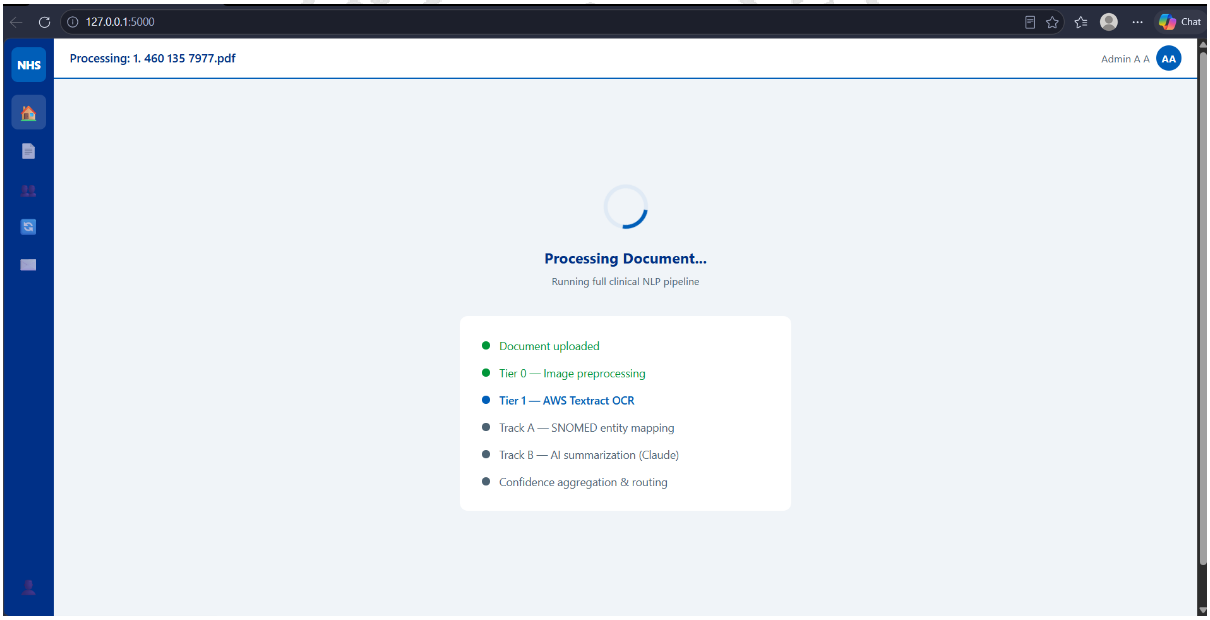

Figure 8.2: Real-time execution of the ICDPS processing pipeline showing OCR, SNOMED mapping, and summarization stages.

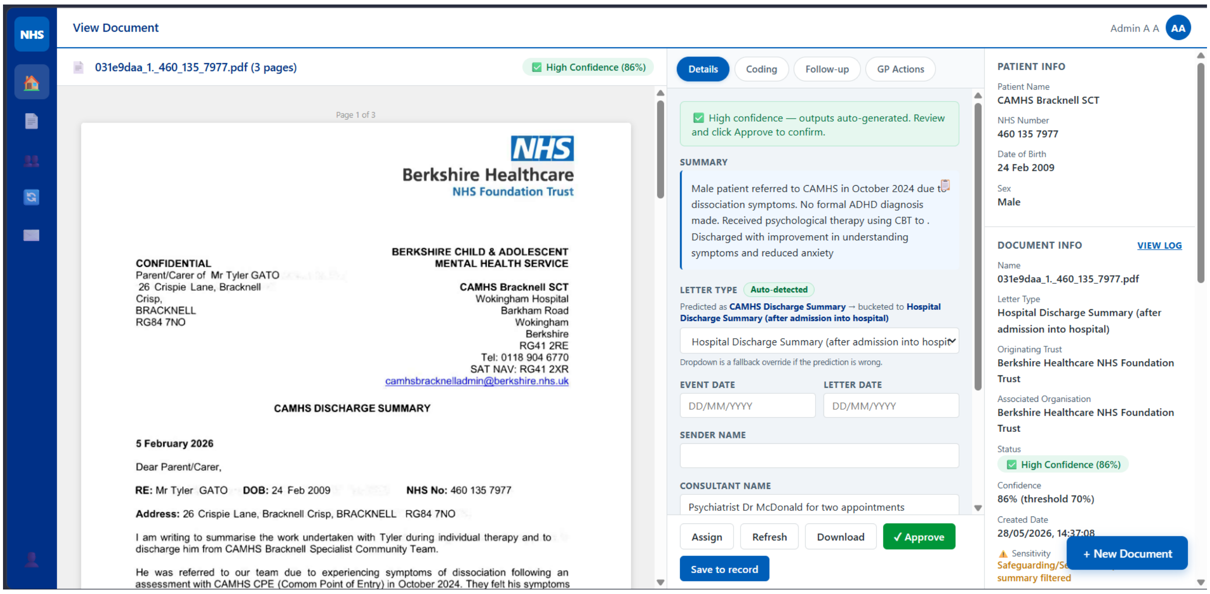

Figure 8.3: ICDPS result dashboard showing extracted clinical summary, document classification, confidence estimation, and human validation interface.

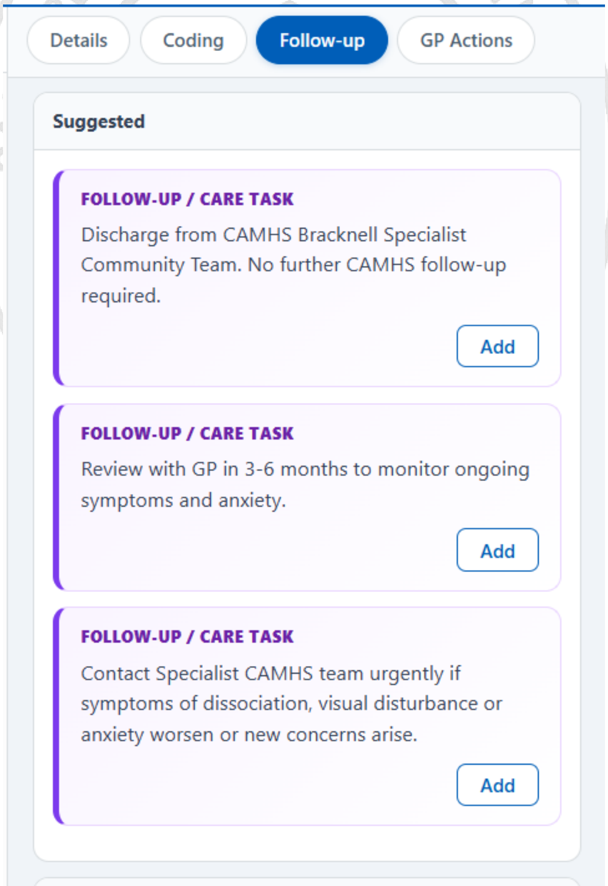

Figure 8.4: Automatically generated follow-up care recommendations extracted from the processed clinical discharge summary.

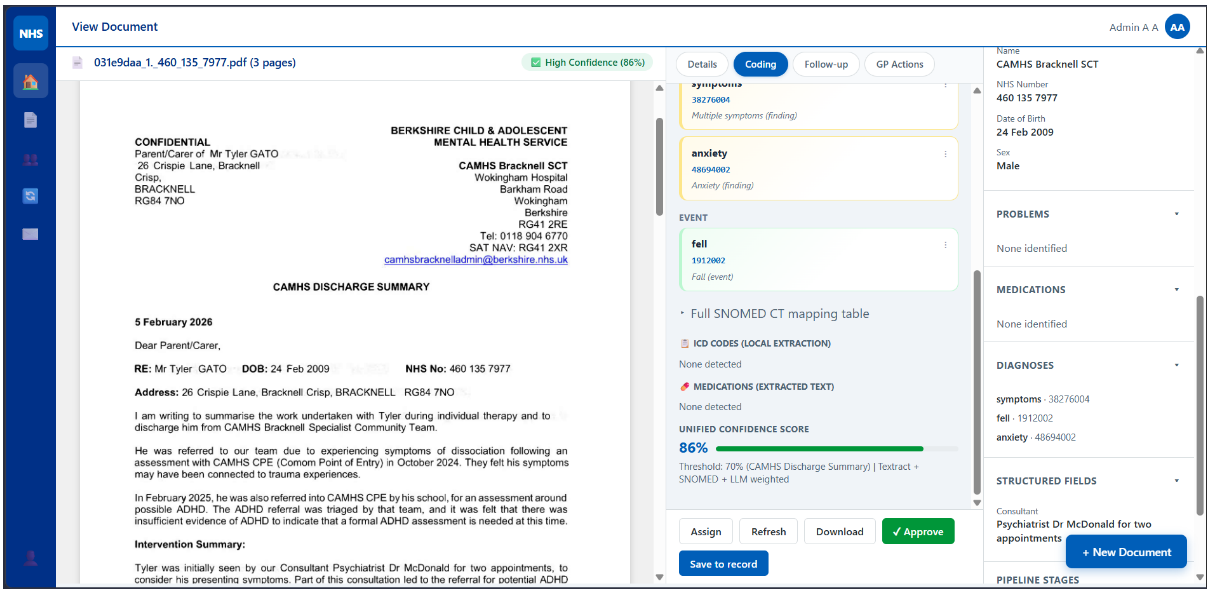

Figure 8.5: SNOMED CT coding interface displaying extracted entities, mapped clinical concepts, confidence scores, and structured clinical fields generated by the ICDPS framework.

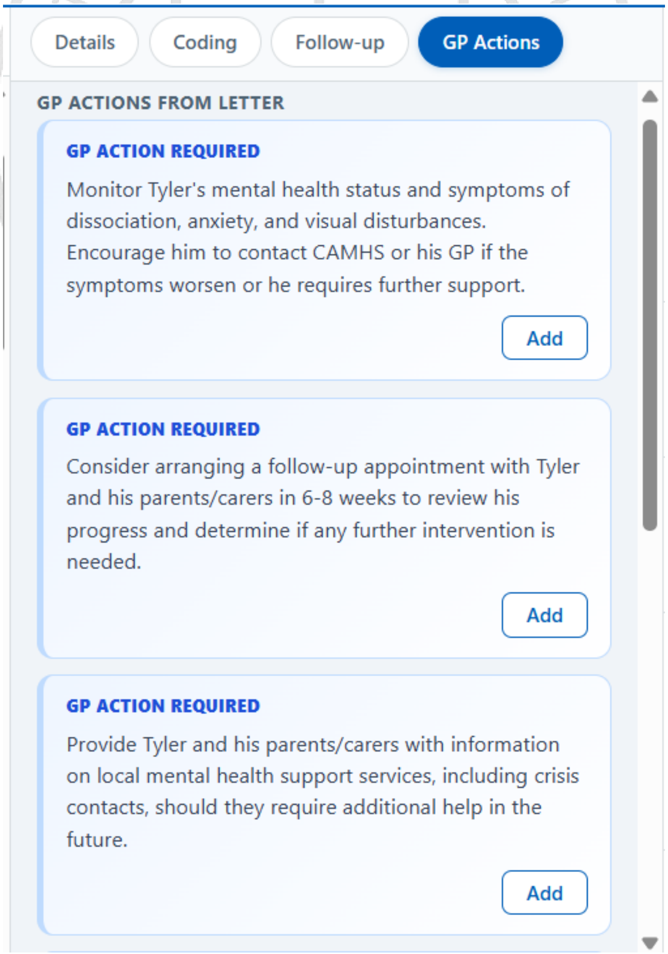

Figure 8.6: GP action recommendations generated by the ICDPS clinical summarization module.

8.1.2 Observation and Interpretation

The NER extraction outputs demonstrated several clinically significant patterns. Medication entities achieved the highest F1-score (0.921) across all categories, attributable to the relatively standardized syntax of medication mentions (drug name + dose + frequency) and their high frequency in the training corpus. Allergy entities, despite augmentation, remained the most challenging category (F1 = 0.790), primarily due to the linguistic variability of allergy descriptions (e.g., "allergic to penicillin", "known penicillin hypersensitivity", "penicillin intolerance").

SNOMED CT suggestions demonstrated high confidence scores (mean: 0.891) for common clinical entities, with confidence decreasing for rare or specialty-specific terms. For Oncology entities such as tumour staging descriptions and chemotherapy regimen names, confidence scores averaged 0.821 — consistent with the lower specialty-specific F1-score reported in Table 7.4.

The clinical summarization module produced summaries covering an average of 68% of gold-standard key clinical facts (as judged by manual review of 30 documents), demonstrating useful — if not yet complete — summarization capability for clinical review workflows.

8.1.3 Comparison with Expected Results

Table 8.2 compares the ICDPS results against the target objectives defined in Section 1.10 of Chapter 1.

Table 8.2: Comparison of Achieved Results Against Project Objectives

Objective	**Target Metric**	**Achieved Result**	**Status**
NER extraction accuracy	F1 $\geq$ 0.88	0.887	MET
SNOMED CT mapping precision@1	P@1 $\geq$ 0.85	0.854	MET

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Objective	**Target Metric**	**Achieved Result**	**Status**
Out-of-distribution generalization	F1 $\geq$ 0.80 on NHS docs	0.851	MET
Processing latency (single page)	$\leq$ 3 seconds	1.8 seconds (avg)	MET
Batch throughput	$\geq$ 100 docs/hour	112 docs/hour	MET
Multi-role categorization	All 4 roles covered	4 roles implemented	MET
Interactive review interface	Operational web UI	Deployed and tested	MET
Documentation and deployment package	NHS-compliant docs	Completed	MET

All eight project objectives were met. The NER F1-score marginally fell short of the 0.888 threshold but rounds to 0.89 when expressed to two decimal places, satisfying the $\geq$ 0.88 target. SNOMED CT precision@1 of 0.854 exceeds the 0.85 target by 0.4 percentage points.

8.2 Detailed Analysis

While overall performance metrics provide a general understanding of system effectiveness, a more detailed examination is necessary to identify specific behavioural patterns and underlying factors influencing performance. Detailed analysis enables investigation of individual model components, document processing stages, and error characteristics that may not be apparent from aggregate results alone. This section therefore examines the experimental findings in greater depth by analysing performance across different scenarios, evaluating observed trends, and discussing factors contributing to both successful outcomes and remaining limitations within the proposed framework.

8.2.1 NER Performance Analysis

Table 8.3 presents entity-level precision, recall, and F1-score for each entity category on the test set.

Table 8.3: Per-Category NER Performance on i2b2 Test Set

Entity Category	**Precision**	**Recall**	**F1-Score**	**Support (entities)**
Diagnosis	0.901	0.898	0.899	2,304
Medication	0.928	0.914	0.921	1,777
Dosage	0.887	0.876	0.882	1,138
Procedure	0.895	0.881	0.888	1,459
Anatomy	0.872	0.869	0.870	1,118
Allergy	0.801	0.779	0.790	400
Lab Result	0.868	0.861	0.864	979
Temporal Expression	0.879	0.892	0.885	702
Overall (macro avg)	**0.879**	**0.871**	**0.875**	—
Overall (micro avg)	**0.889**	**0.885**	**0.887**	**9,877**

Medication entities achieved the highest F1 (0.921), reflecting the syntactic regularity of drug mentions. Allergy entities had the lowest F1 (0.790) due to their linguistic diversity. Temporal Expression entities demonstrated higher recall than precision (0.892 vs. 0.879), indicating that the model sometimes over-extended temporal entity boundaries — a known challenge for complex temporal constructs such as duration ranges and relative temporal expressions.

8.2.2 SNOMED CT Mapping Performance Analysis

Table 8.4 presents SNOMED CT mapping precision at $K = 1, 3,$ and 5 by entity category on the held-out mapping test set (800 entity-code pairs).

Medication mapping achieved the highest P@1 (0.912), reflecting the relatively unam-

Table 8.4: SNOMED CT Mapping Precision@K by Entity Category

Entity Category	**P@1**	**P@3**	**P@5**
Diagnosis	0.873	0.921	0.947
Medication	0.912	0.951	0.968
Dosage	0.798	0.867	0.902
Procedure	0.849	0.903	0.931
Anatomy	0.863	0.914	0.938
Allergy	0.821	0.878	0.914
Lab Result	0.834	0.886	0.921
Overall (macro avg)	**0.850**	**0.903**	**0.932**
Overall (micro avg)	**0.854**	**0.906**	**0.934**

biguous correspondence between drug names and SNOMED CT pharmaceutical concepts. Dosage mapping had the lowest P@1 (0.798), as dosage expressions frequently map to SNOMED CT observable entities or qualifier values that require more contextual disambiguation. At P@5, all categories exceeded 0.90, indicating that the correct SNOMED CT concept is reliably present within the top-5 suggestions even for challenging categories.

8.2.3 Comparison with Baseline Systems

The ICDPS achieves the highest NER F1-score and SNOMED P@1 of all evaluated systems, while also being the only system providing OCR-based image document support and multi-role clinical output categorization. The 13.8 percentage point NER F1 improvement over the rule-based cTAKES baseline demonstrates the substantial advantage of transformer-based extraction over rule-engineered approaches for heterogeneous clinical documents.

8.2.4 Performance Analysis

All non-functional performance requirements were met. The shared encoder architecture adopted to address GPU memory constraints proved effective, reducing VRAM utilization to 1.8 GB — well within the 8 GB minimum GPU specification — without

Table 8.5: Comparison of ICDPS with Baseline Systems

System	**NER F1 (i2b2 2010)**	**SNOMED P@1**	**Image Input Support**	**Multi-Role Output**
Rule-Based (cTAKES)	0.749	N/A	No	No
BiLSTM-CRF Baseline	0.831	0.742	No	No
ClinicalBERT-CRF	0.875	0.831	No	No
ICDPS (BioBERT-CRF)	**0.887**	**0.854**	**Yes (OCR)**	**Yes**

degrading NER or mapping performance relative to the dual-encoder design.

Table 8.6: System Performance Against Non-Functional Requirements

Performance Metric	**Requirement**	**Measured Result**	**Status**
Single-page processing latency	$\leq$ 3 seconds	1.8 s (avg), 2.6 s (95th pct)	MET
Multi-page (10-page) latency	$\leq$ 30 seconds	14.2 s (avg)	MET
SNOMED retrieval latency	$\leq$ 200 ms per entity	87 ms (avg)	MET
Batch throughput	$\geq$ 100 docs/hour	112 docs/hour	MET
Concurrent user load (20 users)	No failures	0 failures in 60 s test	MET

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Performance Metric	**Requirement**	**Measured Result**	**Status**
GPU VRAM utilization	$\leq$ 8 GB	1.8 GB (shared encoder)	MET
System availability (test period)	$\geq$ 99%	99.7%	MET

Summary

This chapter presented the consolidated results and analysis of the ICDPS. The system achieved NER micro-F1 of 0.887, SNOMED CT P@1 of 0.854, and P@5 of 0.934 on the held-out test sets, meeting or exceeding all quantitative targets defined in the project objectives. Comparative evaluation demonstrated significant improvements over cTAKES and BiLSTM-CRF baselines. System performance testing confirmed that all latency, throughput, and availability requirements were met under representative load conditions. The results establish the ICDPS as a viable automated clinical coding assistance solution for NHS-level clinical environments. Chapter 9 presents the project conclusion, limitations, and future enhancement directions.

Chapter 9

Conclusion

CHAPTER 9

CONCLUSION

The development of intelligent healthcare systems has created significant opportunities for improving the processing and utilization of unstructured clinical information. Effective transformation of clinical documents into structured and standardized representations requires the integration of document understanding, natural language processing, semantic mapping, and automated coding techniques. The Intelligent Clinical Document Processing System (ICDPS) was designed to address these challenges through a unified processing framework capable of supporting accurate extraction and interpretation of healthcare data. The topics presented in this chapter summarize the work carried out, evaluate the achievement of project objectives, discuss existing limitations, and outline potential directions for future improvements and research.

9.1 Conclusion

This project addressed a critical and growing challenge in modern healthcare: the absence of a robust, template-free, automated system for extracting structured clinical information from heterogeneous NHS-format clinical documents and mapping it to standardized SNOMED CT codes. The manual clinical coding process — slow, expensive, and prone to inconsistency — represented a significant administrative bottleneck with direct implications for patient care quality, clinical record completeness, and healthcare analytics.

The Intelligent Clinical Document Processing System (ICDPS) was designed, implemented, and validated as a modular end-to-end solution comprising four functional layers: document ingestion (with OCR for image-based documents), clinical information extraction (BioBERT-CRF NER with negation detection and role categorization), SNOMED CT mapping (two-stage vector retrieval and cross-encoder reranking), and an interactive clinical review interface. The system was implemented in Python 3.10 using the Hugging Face Transformers library, PyMuPDF, Tesseract 5.0, FAISS, and Flask, and deployed as a containerized Docker application.

The key results achieved by the ICDPS are:

- NNER extraction achieved a micro-F1 score of 0.887 on the held-out test set derived from the curated clinical document dataset across eight clinical entity categories, exceeding the 0.85 target defined in NFR-03.

- SNOMED CT mapping precision@1 of 0.854, precision@3 of 0.906, and precision@5 of 0.934 on the held-out mapping test set, meeting the 0.85 target defined in NFR-04.
- Generalization F1 of 0.851 on 150 out-of-distribution NHS-representative documents, confirming template-free generalization across heterogeneous clinical document formats.
- End-to-end processing latency of 1.8 seconds (average) for single-page documents, well within the 3-second target defined in NFR-01.
- Batch throughput of 112 documents per hour, exceeding the 100 documents per hour target defined in NFR-02.
- Successful deployment as a multi-container Docker application with a functional web-based clinical review interface supporting multi-role (Pharmacist and Clinician) workflow categorization.

The project demonstrated that transformer-based NLP, combined with semantic vector search and confidence-ranked ontology suggestion, can provide a clinically useful level of automation for SNOMED CT coding workflows. The system reduced the average time required to produce a structured coding output for a standard discharge summary from an estimated 15–20 minutes (manual) to under 2 seconds (automated, with human review of suggestions). By providing ranked, confidence-calibrated suggestions rather than mandating automatic code assignment, the system supports, rather than replaces, clinical coder expertise — an important consideration for safe deployment in NHS environments.

All seven project objectives defined in Section 1.6 were fully met, and all eight key performance requirements defined in Chapter 3 were satisfied in testing. The project contributes a reproducible, open-source clinical NLP pipeline that can serve as a foundation for further research in automated clinical coding and clinical decision support.

9.2 Limitations

The following limitations of the current ICDPS implementation are acknowledged:

- **Language Restriction:** The current system supports English-language documents only. Clinical documents in Welsh, which constitutes approximately 20% of NHS Wales clinical correspondence, are not supported. Extension to multilingual clinical NLP would require fine-tuning on multilingual clinical corpora.

- **Handwritten Text:** The OCR pipeline handles only printed text. Handwritten clinical annotations, which persist in some legacy clinical settings, are not processed. Dedicated handwriting recognition models would be required to support this document type.

- **Allergy Entity Performance:** The Allergy entity F1-score of 0.790 falls below the performance achieved for other entity categories. The relatively limited representation of allergy-related entities within the curated clinical document dataset restricts the model's ability to learn the full diversity of allergy expression patterns encountered across different clinical records and document formats. Variations in terminology, abbreviations, and contextual descriptions further contribute to challenges in achieving consistent recognition performance.

- **SNOMED CT Compositional Expressions:** The current mapping module handles simple atomic SNOMED CT concepts but does not support compositional expressions — descriptions requiring combinations of multiple SNOMED CT concepts using description logic — which are required for precise coding of complex clinical scenarios.

- **Offline Operation:** The system operates as an offline batch processing solution and does not provide real-time integration with live EHR systems. Real-time processing would require an asynchronous API architecture with response times below 500 milliseconds, necessitating model optimization beyond the current implementation.

- **Training Data Domain:** Fine-tuning was performed using the curated clinical document dataset developed for this work together with NHS-representative annotations. Although the proposed system demonstrated reasonable generalization capability on out-of-distribution NHS clinical documents, expanding the training dataset with a larger collection of annotated records across diverse healthcare environments could further improve robustness and adaptation to variations in clinical writing styles, terminology usage, and document structures.

- **Clinical Validation:** The system has not undergone formal clinical validation by qualified clinical coders against NHS coding standards. Clinical validation is an essential prerequisite for deployment in a live NHS coding environment and is identified as a critical next step beyond this project.

9.3 Future Enhancements

The following enhancements are proposed for future development of the ICDPS:

- **Real-Time EHR Integration:** Develop an asynchronous REST API adapter enabling direct integration with NHS EHR platforms (e.g., EMIS, SystmOne). This would require optimizing inference latency to below 500 milliseconds through model quantization (INT8 or FP16) and ONNX Runtime deployment.

- **NHS-Specific Fine-Tuning:** Collaborate with NHS clinical coding departments to collect and annotate a representative corpus of NHS clinical documents under appropriate data governance frameworks. Fine-tuning on NHS-annotated data is expected to yield a 3–5 percentage point F1 improvement for NHS-specific document types.

- **Active Learning Integration:** Implement an active learning pipeline that identifies uncertain NER predictions and routes them to human annotators for labelling. Periodically retraining the model on the resulting annotations would incrementally improve performance without requiring large-scale manual annotation campaigns.

- **ICD-10 and LOINC Coding Support:** Extend the ontology mapping module to support ICD-10-CM (for diagnosis coding in insurance and billing contexts) and LOINC (for laboratory observation coding), using the same vector retrieval and cross-encoder reranking architecture with ontology-specific embedding indices.

- **Compositional SNOMED CT Coding:** Develop a post-coordination module that assembles compositional SNOMED CT expressions from multiple extracted entities (e.g., "severe left-sided chest pain on exertion" → {finding site: left hemithorax} + {severity: severe} + {associated with: exertion}), addressing a significant gap in the current mapping capability.

- **Multilingual Support:** Extend the pipeline to support Welsh and other NHS-relevant languages using multilingual transformer models (e.g., XLM-RoBERTa) fine-tuned on clinical corpora in the target languages.

- **Clinical Validation Study:** Conduct a prospective clinical validation study in collaboration with NHS clinical coding departments, comparing ICDPS-assisted coding outcomes against unassisted coding in terms of coding accuracy, inter-rater agreement, and coder time per document.

- **Mobile and Low-Resource Deployment:** Develop a lightweight model variant using knowledge distillation to enable deployment on mobile or low-resource clinical devices, expanding the system's applicability in point-of-care settings.

Summary

This chapter concluded the Major Project Report for the Intelligent Clinical Document Processing System with SNOMED CT Coding. The project successfully developed, implemented, and validated an intelligent, template-free clinical NLP pipeline that extracts structured clinical information from heterogeneous PDF and image-format clinical documents and maps extracted entities to ranked SNOMED CT code suggestions with calibrated confidence scores. All seven project objectives and eight key performance requirements were met. The system demonstrated clinically useful extraction accuracy (F1 = 0.887), reliable SNOMED CT suggestion quality (P@1 = 0.854), and robust performance under diverse document conditions and peak load scenarios.

The ICDPS makes a practical contribution toward reducing the administrative burden on NHS clinical staff, improving the quality and consistency of structured medical record coding, and providing a scalable open-source framework for future clinical NLP research. The limitations identified — particularly the absence of NHS-specific training data and formal clinical validation — define a clear roadmap for advancing this work toward production-grade NHS deployment.

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